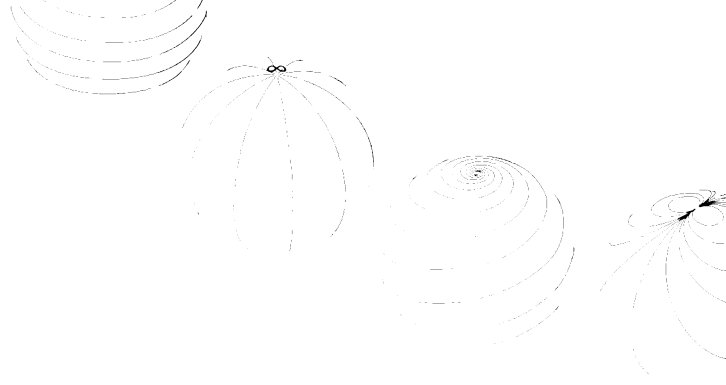




CHAPTER 5

Further Geometry of Differentiation

5.1 Cauchy–Riemann Revealed

5.1.1 Introduction

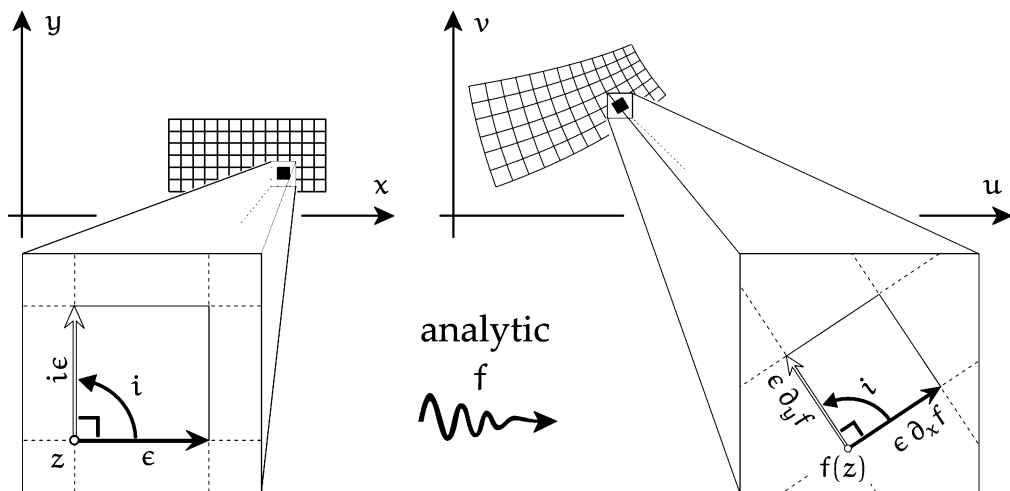
In the previous chapter we began to investigate the remarkable nature of analytic functions in $\mathbb{C}$ by studying mappings in the less structured realm of $\mathbb{R}^2$. In particular, the Jacobian provided us with a painless way of deriving the Cauchy–Riemann characterization of analytic functions, and also of computing their amplitwists. However, this approach was rather indirect.

In this chapter we will instead study differentiation directly in the complex plane, primarily through the use of *ultimate equality*—the powerful geometrical tool that Newton unleashed in his great *Principia* of 1687; see both the original and the new Preface. However, in the original edition these arguments were expressed mainly in the *language* of infinitesimals, and we have left this mode of expression largely intact. But in the *captions*, all of which are new to this edition, we have felt free to employ the $\asymp$ -notation.

Our first application of this approach will be the rederivation of the Cauchy–Riemann (henceforth “CR”) equations, and the discovery of new forms that they can take on.

5.1.2 The Cartesian Form

Consider a very fine mesh of squares aligned with the real and imaginary axes. See the top left of [5.1]. Under an analytic mapping each infinitesimal square will be amplitwisted to produce an image that is also square. We will show that the CR equations are nothing more than a symbolic restatement of this geometric fact.



[5.1] Geometric derivation of the Cartesian Cauchy–Riemann equations. *The image of the real edge ϵ of the small (ultimately vanishing) square on the left is ultimately equal to $\epsilon \partial_x f$ on the right. Likewise, the image of the imaginary edge $i\epsilon$ of the square on the left is ultimately equal to $\epsilon \partial_y f$ on the right. But if f is analytic, then the image of the square on the left must ultimately be another square on the right. So one edge of the image is ultimately equal to the other edge rotated by $(\pi/2)$, that is, multiplied by i . Thus, $\epsilon \partial_y f \asymp i\epsilon \partial_x f \implies \partial_y f = i \partial_x f$, which is the compact Cartesian form of the celebrated Cauchy–Riemann equations.*

Zoom in on an individual square and its image, as depicted in the bottom half of [5.1]. Suppose, as drawn, that the initial square has side ϵ . If we start at z and then move a distance ϵ in the x -direction, the image will move along a complex number given by

$$(\text{change in } x) \cdot (\text{rate of change with } x \text{ of the image } f) = \epsilon \partial_x f.$$

Similarly, if the point moves along the vertical edge by going ϵ in the y -direction, then its image will move along $\epsilon \partial_y f$. Now since these two image vectors span a square they must be related by a simple rotation of $\pi/2$, that is by multiplication with i . After cancelling ϵ , we thus obtain

$$i \partial_x f = \partial_y f, \quad (5.1)$$

et voilà! That this is indeed a compact form of the CR equations may be seen by inserting $f = u + iv$:

$$i \partial_x (u + iv) = \partial_y (u + iv),$$

and then equating real and imaginary parts to yield

$$\partial_x u = \partial_y v \quad \text{and} \quad \partial_x v = -\partial_y u, \quad (5.2)$$

just as before.

To obtain the amplitwist itself, we recall that each infinitesimal arrow is taken to its image by multiplication with f' . Now, since we know what the images are for the two sides of the square, we easily deduce

$$\begin{aligned}\epsilon &\longmapsto \epsilon f' = \epsilon \partial_x f \\ \Rightarrow f' &= \partial_x f\end{aligned}$$

and

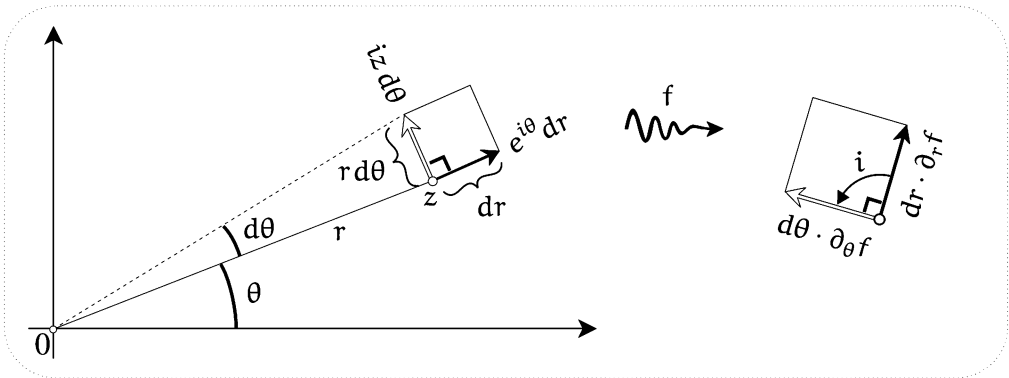
$$\begin{aligned}i\epsilon &\longmapsto i\epsilon f' = \epsilon \partial_y f \\ \Rightarrow f' &= -i \partial_y f.\end{aligned}$$

These are the same formulae (4.8) and (4.9) that we previously obtained, but *now* we understand their geometrical *meaning*.

5.1.3 The Polar Form

Equation (5.2) is the most common way of writing CR, but it isn't the only way. It took this form because we chose to describe both complex planes in terms of their real and imaginary parts, that is with Cartesian coordinates. Thus we could briefly describe (5.2) as being the Cart.–Cart. form. In Ex. 10, p. 240 we retained Cartesian coordinates for the first plane but employed polar coordinates in the image plane; this led to another form (Cart.–Polar) of CR. As the next example of our geometric method we will derive the Polar–Cart. form of the equations.

In order to do this, we begin with an infinitesimal square adapted to polar coordinates. See [5.2]. If we start at z and increase r by dr , then we obtain $e^{i\theta} dr$ as the



[5.2] Geometric derivation of the polar Cauchy–Riemann equations. If z moves radially by dr then its image ultimately moves $dr \cdot \partial_r f$. If the angle of z increases by $d\theta$, ultimately causing z to move $r d\theta$ orthogonally, then its image ultimately moves $d\theta \cdot \partial_\theta f$, as illustrated. If the initial shape is ultimately a square (so that $dr \asymp r d\theta$) then an analytic f will ultimately map it to another square, so $d\theta \cdot \partial_\theta f \asymp i dr \cdot \partial_r f \asymp i r d\theta \cdot \partial_r f$. Thus the Cauchy–Riemann equations take the form $\partial_\theta f = i r \partial_r f$.

radial edge. If, on the other hand, we increase θ by $d\theta$, then the point will move in the perpendicular direction given by $ie^{i\theta}$. As $d\theta$ tends to zero, this edge is ultimately equal to an infinitesimal arc of circle of length $r d\theta$; the complex number describing it will therefore be $ie^{i\theta} r d\theta = iz d\theta$. It's also clear from our picture that

$$\text{initially square} \iff dr = r d\theta. \quad (5.3)$$

Now look at the image. Just as before, if we increase r by dr then the image will move along $dr \cdot \partial_r f$; likewise, changing θ by $d\theta$ will move the image along $d\theta \cdot \partial_\theta f$. If the mapping is analytic then these again span a square, and so the latter must be i times the former:

$$d\theta \cdot \partial_\theta f = i dr \cdot \partial_r f.$$

Substituting (5.3) into this, and cancelling $d\theta$, we obtain

$$\partial_\theta f = ir \partial_r f \quad (5.4)$$

as the new compact form of CR. By inserting $f = u + iv$, the reader may verify that (5.4) is equivalent to the following pair of Polar–Cart. equations:

$$\partial_\theta v = +r \partial_r u \quad (5.5)$$

$$\partial_\theta u = -r \partial_r v. \quad (5.6)$$

By examining the amplitwist that carries each arrow to its image we can also obtain two expressions for the derivative:

$$\begin{aligned} e^{i\theta} dr &\longmapsto e^{i\theta} dr \cdot f' = dr \cdot \partial_r f \\ &\implies f' = e^{-i\theta} \partial_r f \end{aligned} \quad (5.7)$$

and

$$\begin{aligned} iz d\theta &\longmapsto iz d\theta \cdot f' = d\theta \cdot \partial_\theta f \\ &\implies f' = -(i/z) \partial_\theta f. \end{aligned} \quad (5.8)$$

As a simple example let's take $z^3 = r^3 e^{3i\theta}$. From (5.7) we obtain

$$(z^3)' = e^{-i\theta} 3r^2 e^{3i\theta} = 3r^2 e^{2i\theta} = 3z^2,$$

while from (5.8) we obtain

$$(z^3)' = -(i/z) r^3 3ie^{3i\theta} = -(i/z) 3i z^3 = 3z^2.$$

In obtaining the same answer from both these expressions we have also verified that z^3 actually was analytic in the first place.

Of the four possible ways of writing CR, only one now remains to be found, namely the Polar-Polar form. We leave it to the reader to verify that if we write $f = R e^{i\Psi}$ (cf. Ex. 10, p. 240) then CR takes the form

$$\partial_\theta R = -rR \partial_r \Psi \quad \text{and} \quad R \partial_\theta \Psi = r \partial_r R.$$

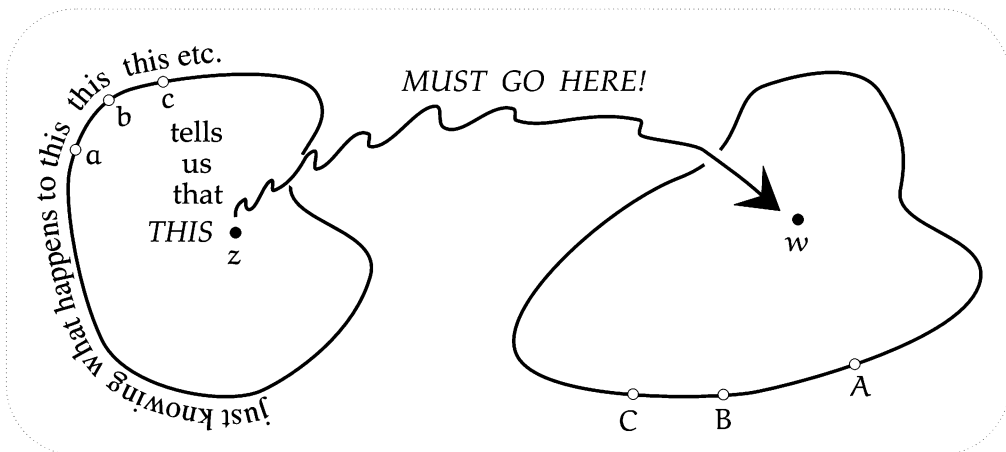
5.2 An Intimation of Rigidity

A recurring theme in complex analysis is the “rigidity” of analytic functions. By this we mean that their highly structured nature (everywhere locally an amplitwist) enables us to pin down their precise behaviour from very limited information. For example, even if we are only told the effect of an analytic function on a small region, then its definition can be extended beyond these confines in a unique way—like a crystal grown from a seed. In fact, given even the meagre knowledge of how an analytic mapping affects a closed curve (just the points *on* the curve, mind you), we can predict precisely what happens to each point *inside*! See [5.3].

Later we will justify these wild claims, and in Chapter 9 we will even find an explicit formula (called *Cauchy’s Formula*) for w in terms of A , B , C , etc. For the moment, though, we will obtain our first glimpse of this rigidity by considering a different kind of partial information.

Consider [5.4]. Origin-centred circles are being mapped to vertical lines, and the larger the circle, the further to the right is the image, but with no restriction on how the lines are spaced. How much information do you think we can gather about an *analytic* mapping possessing this property? Try meditating on this before reading further.

Well, we know that f is conformal and that its local effect is just an amplitwist. Consider the rays emanating from the origin. Since these cut through all the circles at right angles, their images must cut through the vertical lines at right angles, and

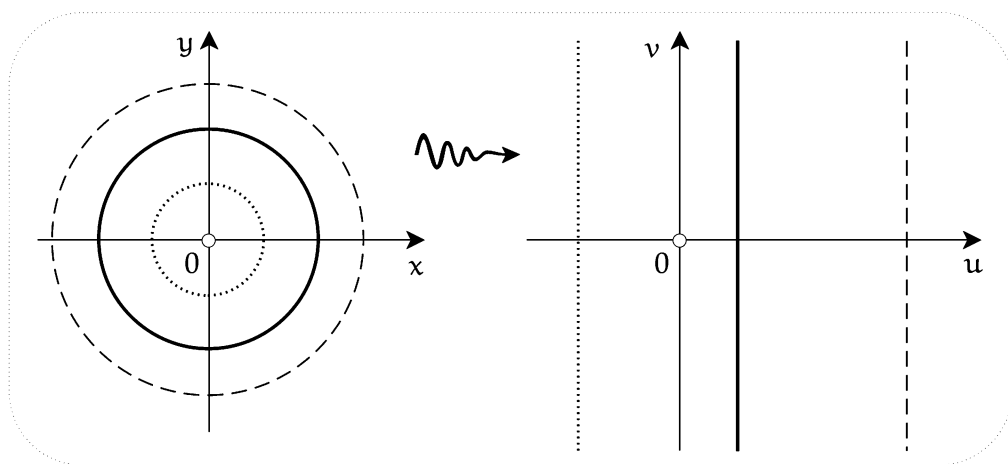


[5.3] Cauchy’s Formula. *The structure of an analytic function is so strong that given even the meagre knowledge of how it affects a closed curve (just the points on the curve, mind you), we can predict precisely what happens to each point inside! The precise recipe is given by Cauchy’s Formula, (9.1), page 485.*

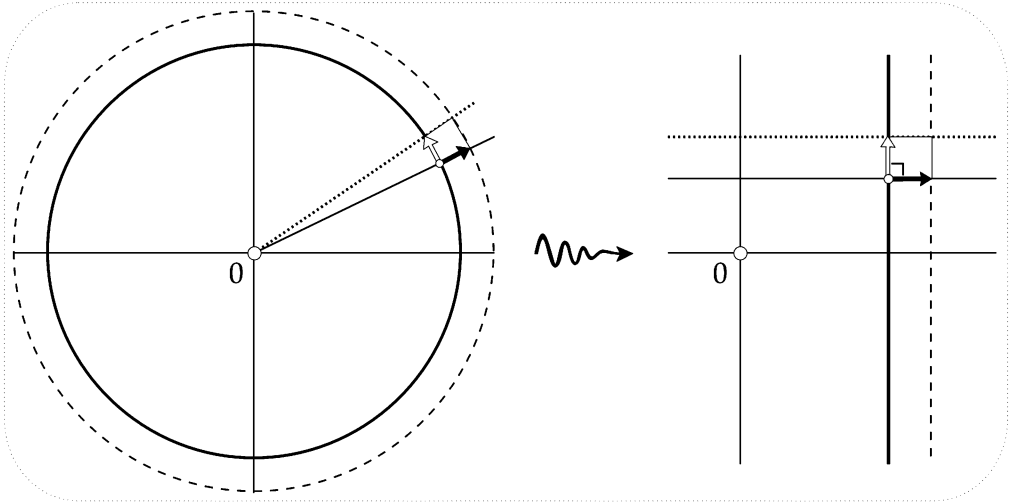
they are thus horizontal lines. In fact, if we swing the ray around counter-clockwise, we can even tell whether its image line will move up or down. Look at [5.5], which depicts the fate of a infinitesimal square bounded by two circles and two rays. We know that the infinitesimal radial arrow connecting the two circles must map to a connecting arrow between the lines going from *left to right*. But since the square is to be amplitwisted, its image must be positioned as shown. Thus we find that a positive rotation of the ray will translate the image line *upwards*.

We have made some good progress, but that we cannot yet have fully captured the consequences of analyticity can be seen from Ex. 5, p. 239. Despite not being analytic, the mapping $(x + iy) \mapsto (x^2 + y^2) + i(y/x)$ was there shown to possess all the above desiderata. Indeed it would be easy to write down an infinitude of nonanalytic functions that would be consistent with the known facts. Very remarkably, and in stark contrast to this, if we restrict ourselves to *analytic* mappings, then we will prove that *there is only one!* To show this we must turn to the CR equations.

In [5.4] we are mapping natural polar objects to natural Cartesian objects, so it's clear that we should employ the Polar–Cart. form, namely (5.5) and (5.6). In order to put them to use, we must first translate [5.4] into ‘Equationspeak’. We could describe the figure by saying that rotating the point only moves the image up and down, not side to side; in other words, varying θ produces no change in u : $\partial_\theta u = 0$. It follows from (5.6) that $\partial_r v = 0$. This says that moving the point radially outwards does not affect the height of the image, and thus that rays are mapped to horizontal



[5.4] There is only one analytic mapping that does this. Suppose that a mapping sends origin-centred circles to vertical lines, and the larger the circle, the further to the right is the image line, but with no restriction on how the lines are spaced. It is easy to write down an infinitude of different non-analytic mappings that do this, but, very remarkably, if we restrict ourselves to analytic mappings, then we can prove that there is only one!



[5.5] Geometric proof of the claim made in the previous figure. *Let us now restrict ourselves to analytic mappings possessing the property in the previous figure. Since rays are orthogonal to origin-centred circles, their images under a conformal mapping must be orthogonal to vertical lines—so the images of rays are horizontal lines. Furthermore, we know that the infinitesimal outward radial arrow connecting the two illustrated circles must map to a connecting arrow between vertical lines going from left to right. Since the square is amplitwisted to its image, we deduce that as the ray rotates counterclockwise, its horizontal image line moves upwards.*

lines. This is old news to us seasoned geometers, but fortunately we have another equation left:

$$\partial_{\theta} V(\theta) = r \partial_r U(r). \quad (5.9)$$

Here we have written $v = V(\theta)$ to stress that it is known to depend only on θ ; similarly for $u = U(r)$.

Now (5.9) looks like an impossible equation, for the LHS quite explicitly depends only on θ , while the RHS is equally emphatic about only depending on r . The only way out of this is for both of these real quantities to equal a constant, say A . Dispensing with the superfluous partial derivatives, we thereby obtain

$$r \frac{dU}{dr} = A \quad \text{and} \quad \frac{dV}{d\theta} = A.$$

Integrating these equations we find that

$$U = A \ln r + \text{const.} \quad \text{and} \quad V = A\theta + \text{const.},$$

and hence

$$U + iV = A (\ln r + i\theta) + B,$$

where $B = \text{const.}$ But we recognize this special combination as none other than the complex logarithm! Thus

$$f(z) = A \log z + B. \quad (5.10)$$

Suppose, more generally, that an analytic function $g(z)$ is known to send circles with centre c to parallel lines making a fixed angle ϕ with the imaginary axis. That there is no fundamental difference between this and the previous case may be seen by considering

$$z \mapsto e^{-i\phi} g(z + c);$$

for you may convince yourself that this possesses property [5.4], and hence it too must equal (5.10). The rigidity of analytic functions has thus led to the rather remarkable conclusion that the complex logarithm is uniquely defined (up to constants) as *the* conformal mapping sending concentric circles to parallel lines.

5.3 Visual Differentiation of $\log(z)$

A fringe benefit of the previous section was the discovery that $\log(z)$ actually is analytic. Since this multifunction finds its simplest representation in Polar–Cart. form, namely

$$\log z = \ln r + i(\theta + 2m\pi),$$

we can easily find its derivative using (5.7) or (5.8). For purposes of illustration, we will now use them both:

$$(\log z)' = e^{-i\theta} \partial_r \log z = e^{-i\theta} (1/r) = 1/z,$$

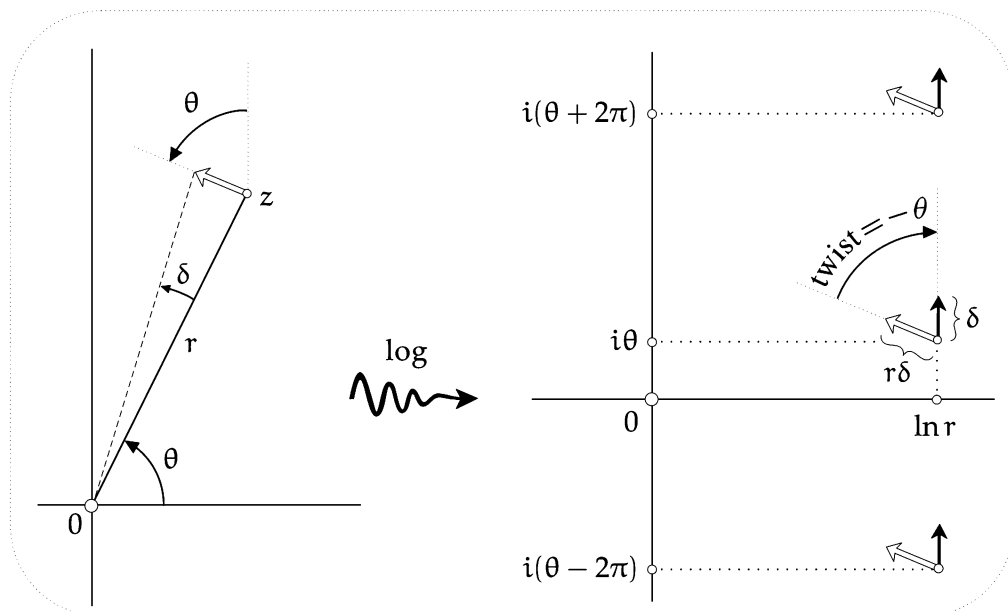
and

$$(\log z)' = -(i/z) \partial_\theta \log z = -(i/z) i = 1/z. \quad (5.11)$$

You notice, of course, how this is formally identical to the case of the ordinary, real logarithm.

You may be wondering how our previous discussion of the branches of this multifunction affects all this. For example, it's interesting how m (which labels the different branches) does not appear in the result (5.11). The basic philosophy of this book is that while it often takes more imagination and effort to find a picture than to do a calculation, the picture will always reward you by bringing you nearer to the Truth. In this spirit, we now find a visual explanation of (5.11) that will also make it clear that the answer does not depend on m .

Equations (5.7) and (5.8) were derived by examining the infinitesimal geometry of a general analytic mapping. Why not then apply this idea to the geometry of a *specific* mapping, and thereby evaluate its amplitwist directly?



[5.6] Geometric demonstration that $[\log(z)]' = (1/z)$. Rotating z by the small, ultimately vanishing angle δ moves it along the white complex number, which ultimately has length $r\delta$. All of the infinitely many images under $\log(z)$ move upwards together, along $i\delta$. Thus, for all branches, the amplification $= (1/r)$, the twist $= -\theta$, and therefore the amplitwist $= (1/r)e^{-i\theta} = (1/z)$.

Consider [5.6], which shows a typical point z and a few of its infinitely many images under $\log$. In order to find the amplitwist we need only find the image of a single arrow emanating from z . The easiest one to find is shown in [5.6], namely an arrow perpendicular to z . Notice how if z makes an angle θ with the horizontal, then the perpendicular vector will make an angle θ with the vertical. Also, if it subtends an infinitesimal angle δ at the origin, then—because it is like a small arc of a circle—its length will be $r\delta$. Now look at the images of z . Since we have purely rotated z , its images will all move vertically up through a distance equal to the angle of rotation δ . To make it easier to see what amplitwist carries the arrow at z into its image, we have drawn copies of the original arrow at each image point. It is now evident from the picture that

$$\begin{aligned} \text{amplification} &= 1/r \\ \text{twist} &= -\theta \\ \implies \text{amplitwist} &= (1/r)e^{-i\theta} = 1/z. \end{aligned}$$

Although all the image vectors emanate from different points in the different branches, they are all identical as vectors, and so it is clear that the amplitwist does not depend on which branch we look at.

5.4 Rules of Differentiation

We already know how to differentiate z^2 and also $\log z$, so how would you use this knowledge to find, for example, the derivative of $\log(z^2 \log z)$? Your immediate reaction (chain and product rules) is quite correct, and in this section we merely verify that all the familiar rules of real differentiation carry over into the complex realm without any changes, at least in appearance.

5.4.1 Composition

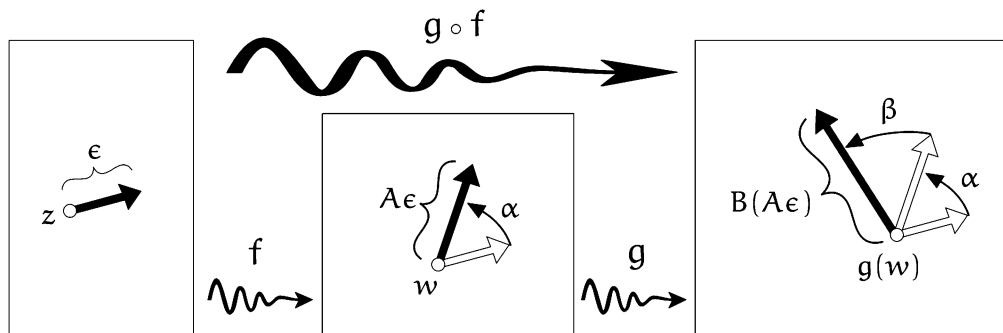
The composite function $(g \circ f)(z) = g[f(z)]$ of course just means “do f , then do g ”. If both f and g are analytic then each of these two steps conserves angles, and therefore the composite mapping does too. We deduce that $g[f(z)]$ is analytic, and we now show that the net amplitwist it produces is correctly given by the chain rule.

Let $f'(z) = A e^{i\alpha}$ and $g'(w) = B e^{i\beta}$, where $w = f(z)$. Consider [5.7]. An infinitesimal arrow at z is amplitwisted by f to produce an image at w ; then this, in its turn, is amplitwisted by g to produce the final image at $g(w)$. It is clear from the picture that

$$\begin{aligned} \text{net amplification} &= AB \\ \text{net twist} &= \alpha + \beta \\ \Rightarrow \text{net amplitwist} &= AB e^{i(\alpha+\beta)}, \end{aligned}$$

and thus we obtain the familiar chain rule:

$$\{g[f(z)]\}' = g'(w) \cdot f'(z). \quad (5.12)$$



[5.7] Geometry of the Chain Rule. The local effect of the composition of two analytic mappings is an amplitwist that is the composition of their separate amplitwists, so we obtain the familiar chain rule.

As an example of this we may put $g(z) = kz$. In the last chapter we showed that $g'(z) = k$, and so we now conclude from (5.12) that

$$[kf(z)]' = k f'(z).$$

5.4.2 Inverse Functions

Provided we are not at a critical point (where the derivative vanishes), an infinitesimal disc at z will be amplitwisted to produce an image disc at $w = f(z)$, and these two discs will be in one-to-one correspondence. See [5.8]. An analytic function thus always possesses a local inverse in this sense, and we wish to know its derivative.

Clearly, the amplitwist that returns the image disc to its original state has reciprocal amplification, and opposite twist:

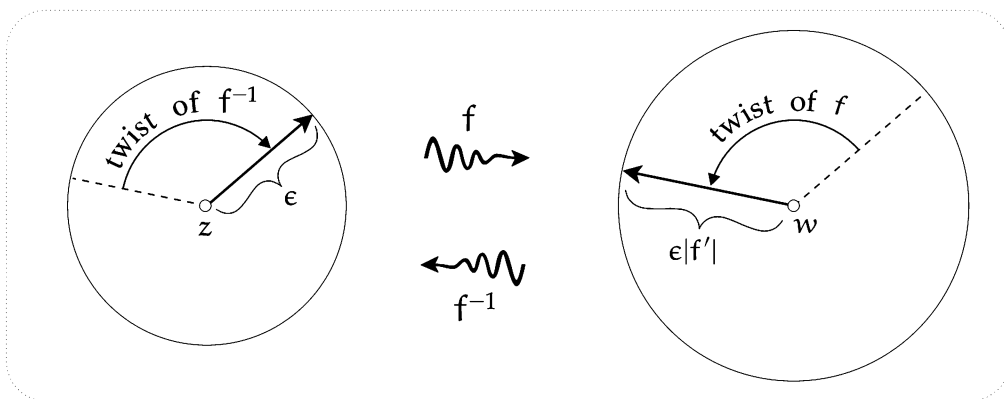
$$\begin{aligned} \text{amplification of } f^{-1} \text{ at } w &= 1/(\text{amplification of } f \text{ at } z) = 1/|f'(z)| \\ \text{twist of } f^{-1} \text{ at } w &= -(\text{twist of } f \text{ at } z) = \arg[1/f'(z)] \\ \implies [f^{-1}(w)]' &= 1/f'(z). \end{aligned} \quad (5.13)$$

By way of example, consider $w = f(z) = \log z$, for which $z = f^{-1}(w) = e^w$. From (5.13) we find that

$$(e^w)' = 1/(\log z)' = z = e^w, \quad (5.14)$$

in agreement with your calculation in Ex. 13, p. 240. Later we will give a visual derivation of (5.14).

Both (5.12) and (5.13) could have been derived even more quickly if we had directly employed the algebraic idea of the image arrow being f' times the original one. We chose instead to keep the geometry to the fore, and reserved the algebra of multiplication for the final encoding of the results as (5.12) and (5.13). However, to derive the next two rules by pure geometry would be cumbersome, so we will use a little algebra.



[5.8] The amplitwist of the inverse of an analytic mapping has the opposite twist and the reciprocal amplification, and so we obtain the familiar rule $[f^{-1}(w)]' = 1/f'(z)$.

5.4.3 Addition and Multiplication

On the far left of [5.9] we see an infinitesimal arrow ξ connecting z to a neighbouring point. The images of these two points under f and (separately) g , are shown in the middle of the figure. Lastly, we either add or multiply these points to obtain the two points on the far right. By examining the image vector connecting these final points we can deduce the amplitwists of $(f + g)$ and fg , respectively. In [5.9] we see that

$$A - a \asymp \xi f' \quad \text{and} \quad B - b \asymp \xi g',$$

so that

$$(A + B) - (a + b) \asymp \underbrace{\xi(f' + g')}_{\text{ultimate image of } \xi},$$

and hence we obtain the addition rule:

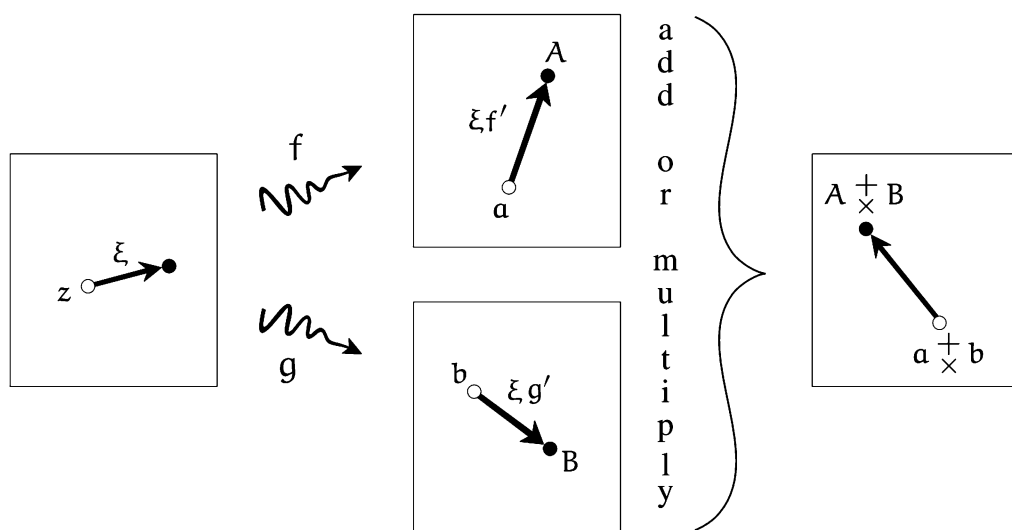
$$(f + g)' = f' + g'. \quad (5.15)$$

Likewise, we find that

$$AB - ab \asymp \underbrace{\xi(f' b + a g')}_{\text{ultimate image of } \xi},$$

and thereby obtain the product rule:

$$(fg)' = f'g + fg'. \quad (5.16)$$



[5.9] If f and g are analytic, then by considering the image of the small, ultimately vanishing complex number ξ , under first their sum and then their product, we deduce that $(f + g)' = f' + g'$, and $(fg)' = f'g + fg'$, just as in elementary real calculus.

5.5 Polynomials, Power Series, and Rational Functions

5.5.1 Polynomials

We can look at the rules of the previous section from a slightly different point of view. Take rule (5.16), for example. In a way, what is on the RHS is less important than the fact that there *is* a RHS. By this we mean that we have here a recipe for creating new analytic functions: ‘given two such functions, form their product’. Likewise, each of our other rules can be thought of as a means of producing new analytic functions from old. The analytic functions are indeed the aristocrats of the complex plane, but provided they only mate with their own kind, and only in ways sanctioned by the rules (which allow many forms of incest!), their offspring will also be aristocrats. For example, suppose we start with only the mapping $z \mapsto z$, which is known to be analytic. Our rules now quickly generate $z^2, z^3, \dots$, and thence any polynomial.

Consider a typical polynomial of degree n :

$$S_n(z) = a_0 + a_1 z + a_2 z^2 + \dots + a_n z^n.$$

We have just seen that this is analytic, and thus it maps an infinitesimal disc at p to another at $S_n(p)$. Furthermore, the amplitwist that transforms the former into the latter is, according to (5.15),

$$S'_n(z) = (a_0)' + (a_1 z)' + (a_2 z^2)' + \dots + (a_n z^n)'.$$

We already know how to differentiate the first four terms, and in the next section we will confirm that in general $(z^m)' = m z^{m-1}$, as you no doubt anticipated. Thus

$$S'_n(z) = a_1 + 2a_2 z + 3a_3 z^2 + \dots + n a_n z^{n-1}. \quad (5.17)$$

5.5.2 Power Series

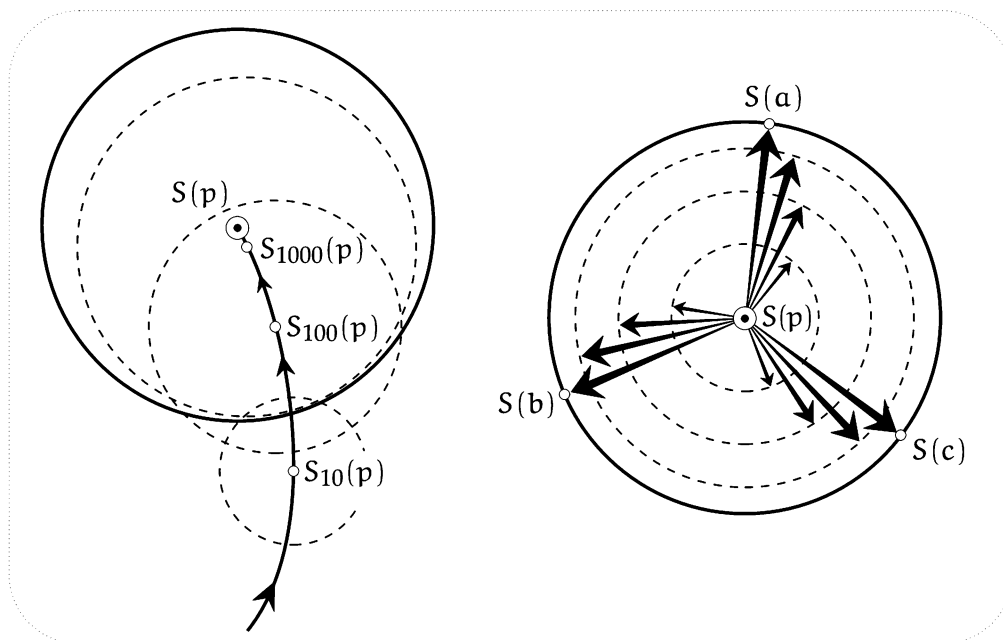
This discussion of polynomials naturally leads to the investigation of power series. In Chapter 2 we discussed how a convergent power series¹

$$S(z) = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \dots \quad (5.18)$$

could be approximated by a polynomial S_n . We explained how the effect of S within its circle of convergence could be mimicked by S_n , with arbitrarily high accuracy, simply by taking a sufficiently high value of n .

Of course the question we now face is whether power series are analytic, and if they are, how are we to calculate their derivatives? We will see that the answers to these questions are “yes” and “(5.17)”.

¹ For simplicity’s sake we shall use a power series centred at the origin. However, as we pointed out in Chapter 2, this does not involve any loss of generality.



[5.10] Power series are analytic. *On the left, we see the images of a small, ultimately vanishing disc D centred at p under the higher and higher degree polynomial approximations $S_n(z)$ of a power series, $S(z)$. On the right, we see these same image discs, now translated to have their common centre at $S(p)$. All these polynomial approximations are analytic, so three equally spaced points on the rim of D will be amplitwisted to their images on the rims of the image discs, and so it becomes clear that $S(z)$ is indeed analytic, and that its amplitwist is the limit of the amplitwists of S_n as n goes to infinity. [For ease of visualization, here the amplification and twist are both depicted as monotonically increasing with n , but this will not be the case in general.]*

Consider an infinitesimal disc D with centre p . If p is inside the circle of convergence of S , then so is a sufficiently small D . The series (5.18) therefore converges at all points of D , and thus S maps the disc to some infinitesimal unknown shape $S(D)$ covering $S(p)$. Now look at the left of [5.10]. This shows a magnified view of the successive images of D [itself not shown in the figure] under S_{10} , S_{100} , S_{1000} , etc. Since each of these polynomials is known to be analytic, each image is a disc. However, it is also known that these images will coincide, ever more perfectly, with $S(D)$. Thus S sends infinitesimal discs to other discs, and it is therefore analytic.

We have tried to make this plainer still on the right of [5.10]. Since we are now only interested in amplitwists, the actual image points are unimportant—we only care about the connecting arrows between them. To make it easier to watch what is happening to these arrows we have translated the discs—which doesn't affect the vectors—so that their centres all coincide at $S(p)$. By way of illustration, we now consider the fate, as n increases, of three equally spaced vectors from p to three equally spaced points (a, b, c) on the rim of D . Each of the analytic mappings S_n

amplitwists these vectors to three equally spaced image vectors. The figure shows the gradual evolution² of these images towards their final state (given by S) as we successively apply S_{10} , S_{100} , etc. The amplitwist that carries the arrows of D into these images therefore undergoes a corresponding evolution towards a final value. The amplitwist S' that carries the original vectors of D to their ultimate images is thus mimicked with arbitrarily high accuracy by S'_n , as n increases. Therefore

$$S'(z) = a_1 + 2a_2 z + 3a_3 z^2 + 4a_4 z^3 + \cdots \quad (5.19)$$

We have reached a very important conclusion. Any power series is analytic within its disc of convergence, and its derivative is obtained simply by differentiating the series term by term. Since the result of this process (5.19) is yet another convergent power series with the same radius of convergence, there is nothing to stop us differentiating again. Continuing in this manner, we discover that a power series is infinitely differentiable within its disc of convergence. The reason this is so important is that we will be able to show later that *every* analytic function can be represented locally as a power series, and thus *analytic functions are infinitely differentiable*.

This result is in sharp contrast to the case of real functions. For example, the mileage displayed on the dash of your car is a differentiable function of the time displayed on the clock. In fact the derivative is itself displayed on the speedometer. However, in the instant that you hit the brakes, the second derivative (acceleration) does not exist. More generally, consider the real function that vanishes for negative x , and that equals x^m for non-negative x . This is differentiable $(m - 1)$ times everywhere, but not m times at the origin. Our complex aristocrats will be shown to be quite incapable of stooping to this sort of behaviour.

5.5.3 Rational Functions

Earlier we established that the product rule applies to complex analytic functions, but we neglected to check the quotient rule. We invite you to verify this now, using the same kind of reasoning that led to (5.16). If you get stuck, there is a hint in Ex. 9. In any event, the important point is that the quotient of two analytic functions is also analytic except at the points where it has singularities. In particular, if we apply this result to polynomials then we can conclude that the rational functions are analytic.

The fact that the quotient of two analytic functions is again analytic can be looked at in a rather more geometric way. Let $I(z) = (1/z)$ be the complex inversion mapping. As we discussed at such length in Chapter 3, $I(z)$ is conformal, and hence it is analytic. It follows that if $g(z)$ is analytic, then so is $[1/g(z)]$, because this is the

² For ease of visualization, we have taken both the amplification and the twist to be steadily increasing with n . In general they could exhibit damped oscillations as they settled down to their final values.

composition $(I \circ g)$ of two analytic functions. Finally, if $f(z)$ is analytic, the product rule tells us that $f(z) \cdot [1/g(z)] = [f(z)/g(z)]$ is too.

5.6 Visual Differentiation of the Power Function

We saw in the last section that $z^2, z^3, z^4, \dots$ were all analytic. Composing with complex inversion, it follows that $z^{-2}, z^{-3}, z^{-4}, \dots$ are too. Since the inverse functions (in the sense of [5.8]) are branches of the multifunctions $z^{\pm 1/2}, z^{\pm 1/3}, \dots$ discussed in Chapter 2, it follows that these too are analytic. Composing z^p with $z^{1/q}$ (p, q integers), it follows that any rational power is analytic. Furthermore, since the geometric effect of any real power can be reproduced with arbitrary accuracy by rational powers, it follows that these real powers are also analytic.

The calculation of the derivative of a real power z^a is similar to the example z^3 given on p. 248. We find that

$$(z^a)' = a z^{a-1}, \quad (5.20)$$

just as in ordinary calculus. In fact the real formula $(x^a)' = a x^{a-1}$ can be thought of as the specialization of (5.20) that results when both z and the infinitesimal arrow emanating from it are taken to be on the real axis (cf. [4.7], p. 221).

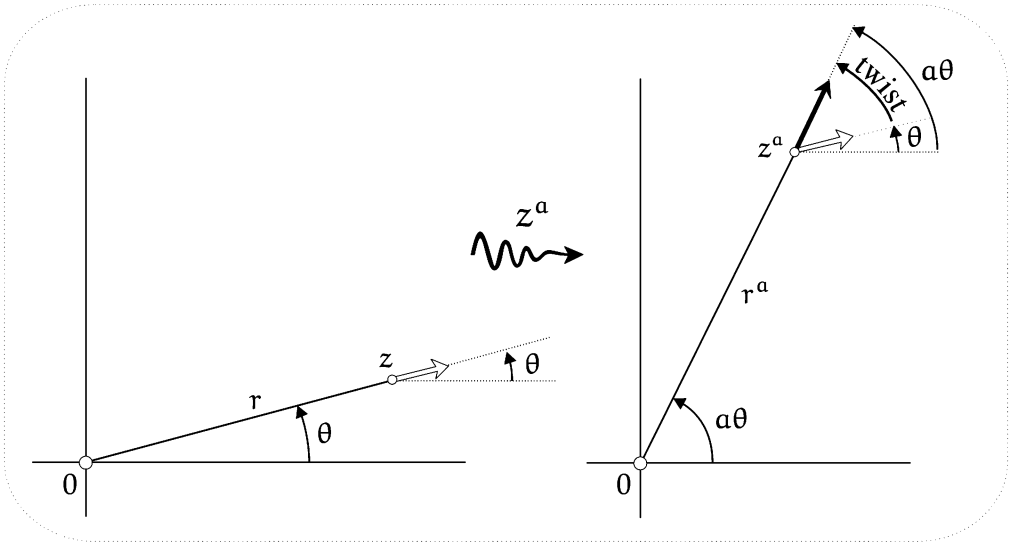
Just as in the case of the complex logarithm, we do not rest at the result (5.20) of a calculation, but rather we stalk the thing to its geometric lair. Since the amplifwist is the same for all arrows, we need only find the image of a single arrow in the direction of our choice. As a first (ill-fated) attempt, consider [5.11], in which we have chosen an arrow parallel to z . To facilitate comparison, we have drawn a copy of the initial arrow at the image point. You can see from the picture that

$$\text{twist} = (a - 1)\theta \quad \text{BUT} \quad \text{amplification} = \text{??????}$$

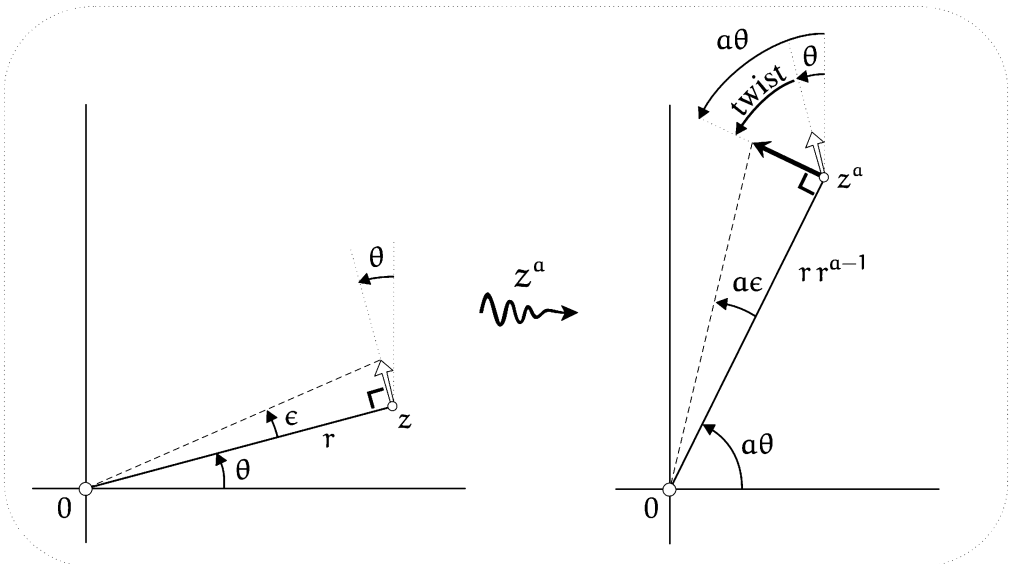
We are thus half-thwarted, for we cannot see how long the image arrow is. In fact to figure this out would require precisely the same calculation (general Binomial Theorem) as is needed in the real case. Oh well, "If at first you don't succeed, ..."

"Try, try an arrow *perpendicular* to z !" From [5.12], we see that this arrow originally makes an angle θ with the vertical, and so after magnifying the angle of z by a , it will make an angle $a\theta$ with the vertical. Once again we see that the twist $= (a - 1)\theta$. However, this time we *can* see the amplification, simply by recognizing that each arrow is an infinitesimal arc of a circle. The angle subtended by the arc has been magnified by a , while the radius of the circle has been magnified by r^{a-1} . The net amplification of the arc is therefore $a r^{a-1}$. Thus

$$\begin{aligned} \text{amplification} &= a r^{a-1} \\ \text{twist} &= (a - 1)\theta \\ \Rightarrow \text{amplitwist} &= a r^{a-1} e^{i(a-1)\theta} = a z^{a-1}. \end{aligned}$$



[5.11] First (ill-fated) attempt to geometrically evaluate the amplitwist of $z \mapsto z^a$. An infinitesimal radial complex number emanating from $z = r e^{i\theta}$ is amplitwisted to a radial complex number emanating from $z = r^a e^{ia\theta}$, and therefore the twist = $(a - 1)\theta$. However, the amplification cannot be geometrically determined from this figure.



[5.12] Second (successful!) attempt to geometrically evaluate the amplitwist of $z \mapsto z^a$. If we instead consider an infinitesimal complex number orthogonal to $z = r e^{i\theta}$, then it is amplitwisted to an infinitesimal complex number orthogonal to $z = r^a e^{ia\theta}$. Therefore, once again, we see that the twist = $(a - 1)\theta$. However, this time the amplification can be geometrically determined: the initial complex number has length $\propto r\epsilon$, while its image has length $\propto r^a(\epsilon)$, so the amplification = $a r^{a-1}$. Therefore, the amplitwist = $(z^a)' = a r^{a-1} e^{i(a-1)\theta} = a z^{a-1}$.

In the above figures $\alpha = 3$, and so there is no ambiguity in the meaning of z^α or $z^{\alpha-1}$. But if α is a fraction, for example, then both z^α and $z^{\alpha-1}$ are multifunctions possessing many different branches. We urge you to redraw [5.12] in such a case. For example, if $\alpha = (1/3)$ then the infinitesimal arrow on the left will have three images on the right, one for each branch of the cube root function. Unlike the case of the multifunction $\log(z)$ (illustrated in [5.6]) these images are obtained by amplitwisting the original arrow by three *different* amounts: each branch of z^α has a different amplitwist. However, your figure will show you [exercise] that

$$\begin{aligned} \text{The amplitwist of each branch of } z^\alpha \text{ is given by } (z^\alpha)' &= \alpha z^\alpha / z, \text{ provided} \\ \text{that the same branch of } z^\alpha \text{ is used on both sides of the equation.} \end{aligned} \quad (5.21)$$

To the best of our knowledge there is no³ direct, intuitive way of understanding the real result $(x^\alpha)' = \alpha x^{\alpha-1}$. It is therefore particularly pleasing that with the greater generality of the complex result (5.21) comes the richer geometry of [5.12] needed to see its truth.

5.7 Visual Differentiation of $\exp(z)$

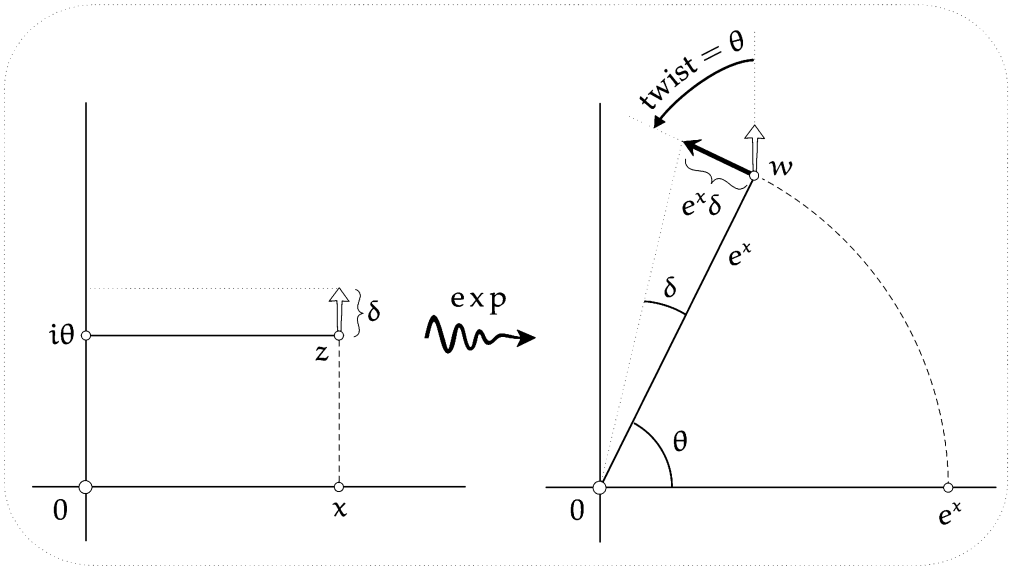
We have already seen that $(e^z)' = e^z$ by calculation, and we will now explain it geometrically. In [5.13] we have written a typical point $z = x + i\theta$ to make it easier to remember that $w = e^z = e^x e^{i\theta}$ has angle θ . Moving z vertically up through a distance δ will rotate the image through an angle δ . Being an infinitesimal arc of circle of radius e^x , the image vector has length $e^x \delta$; its direction is θ to the vertical. As usual, we have copied the original arrow at the image so that we may more clearly see the amplitwist:

$$\begin{aligned} \text{amplification} &= e^x \\ \text{twist} &= \theta \\ \implies \text{amplitwist} &= e^x e^{i\theta} = e^z. \end{aligned}$$

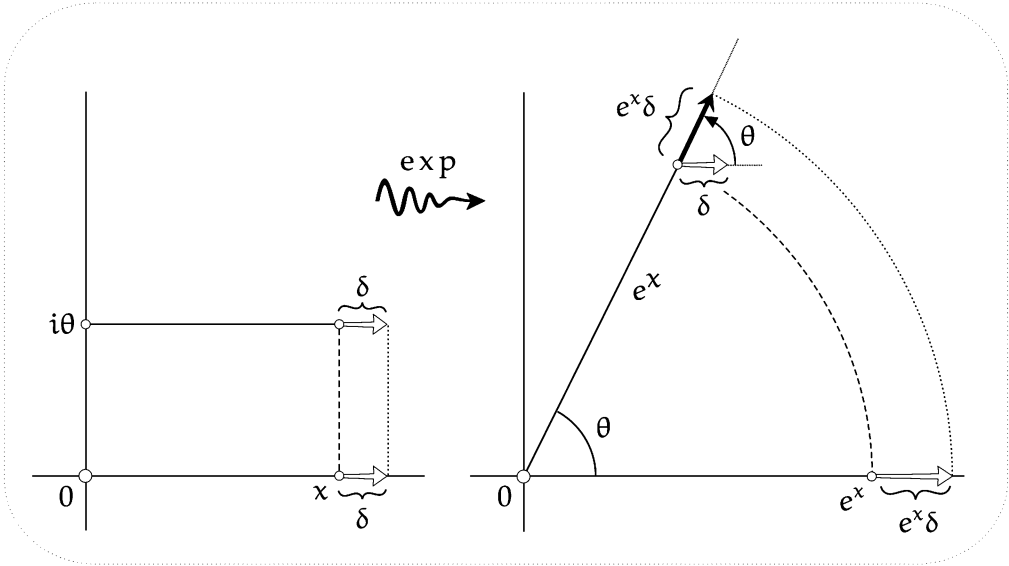
Actually, we have been a little hasty. We haven't really shown yet (at least not geometrically) that e^z is analytic: we don't know if all arrows undergo an equal amplitwist. Figure [5.13] tells us that *if* it's analytic, then $(e^z)' = e^z$. To establish analyticity we need only see that one other arrow is affected in the same way.

In [5.14] we move z an infinitesimal distance δ in the x -direction, thereby moving the image radially outwards. Now, from ordinary calculus, the amplification produced by e^x along the real axis is e^x (cf. [4.6], p. 220), so the length of this image

³ In special cases there are ways. For example, consider a cube of side x . It is easy to visualize that if we increase the separation of one of the three pairs of faces by δ , we add a layer of volume $x^2 \delta$. The result $(x^3)' = 3x^2$ follows.



[5.13] Geometrical evaluation the amplitwist of $z \mapsto e^z$. An infinitesimal vertical complex number $i\delta$ emanating from $z = x + i\theta$ is amplitwisted to a complex number orthogonal to $w = e^z = e^x e^{i\theta}$ whose length is $\asymp e^x \delta$. Therefore the amplification is e^x and the twist $= \theta$, so the amplitwist $= (e^z)' = e^x e^{i\theta} = e^z$.



[5.14] Verification that $z \mapsto e^z$ is locally an amplitwist. If we instead look at the image of an infinitesimal horizontal complex number emanating from z , we see that it, too, undergoes the same amplitwist as did the vertical one in the previous figure, confirming that all infinitesimal complex numbers undergo the same amplitwist.

vector is $e^x \delta$. It is now clear that this new arrow in [5.14] has indeed undergone precisely the same amplification and twist as that in [5.13], thus establishing the analyticity of e^z .

5.8 Geometric Solution of $E' = E$

Up to now we have motivated the definition of the exponential mapping in rather *ad hoc* ways. We are now in a position to do so in a logically more satisfying manner, although *the* most compelling explanation will have to wait till later.

Consider first the ordinary real function that we write as e^x . As we discussed in Chapter 2, one way of characterizing this function is to say that the slope of its graph is always equal to its height. An equivalent dynamic interpretation would be that if the distance of a particle at time t is e^t , then its speed equals its distance from us. In either event, this amounts to saying that the function satisfies the differential equation

$$E' = E. \quad (5.22)$$

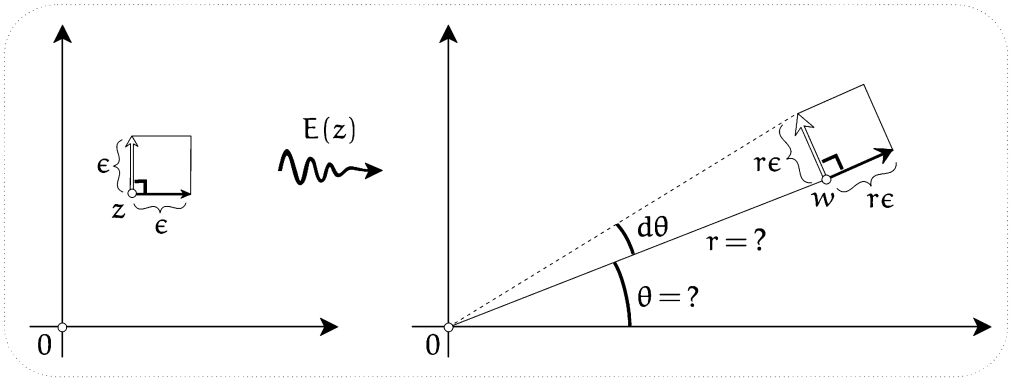
Of course this doesn't quite pin it down since ke^x also obeys (5.22); however, if we insist that the real solution of (5.22) also satisfy $E(0) = 1$, then no ambiguity remains.

The object of this section is to show that the complex exponential function can be characterized in exactly the same way. If a complex-analytic function $E(z)$ is to generalize e^x then it must satisfy (5.22) on the real axis. We will now show geometrically that (5.22) uniquely propagates e^x off the real axis into the plane to produce the familiar complex exponential mapping. The plan will be essentially to reverse the flow of logic associated with [5.13] and [5.14].

A typical point z is being mapped to an unknown image w , where $w = E(z)$ is subject to (5.22). Decoding this equation, we find that it says that vectors emanating from z undergo an amplitwist equal to the image point w . From this alone we will figure out where w must be! In what follows, try to free your mind from assumptions based on your previous knowledge of e^z .

Consider what happens to the little (ultimately vanishing) square of side ϵ shown in [5.15]. Because it's twisted by the angle of w , its horizontal edge becomes parallel to w , while its vertical edge becomes orthogonal to w . Thus horizontal movement of z results in radial movement of the image, while vertical movement results in rotation of the image. The question that now remains is exactly how swift these radial and rotational motions are. Having used the twist, we now turn to the amplification.

If z moves at unit speed in the x -direction, then since the amplification is r , the image moves radially with speed equal to its distance from the origin. But this



[5.15] Geometric solution of $E' = E$. The small, ultimately vanishing square is amplitwisted by $E'(z) = E(z) = w$, so the image of the horizontal edge is radial, and the sides are amplified by $r = |w|$. Thus $d\theta \propto \epsilon$: vertical movement of z produces a numerically equal rotation of w . If z moves at unit speed in the x -direction, then since the amplification is r , the image moves radially with speed equal to its distance from the origin. But this is just the familiar property of the ordinary exponential function. Thus E maps horizontal lines exponentially onto rays.

is just the familiar property of the ordinary exponential function. Thus E maps horizontal lines exponentially onto rays. If we now insist that $E(0) = 1$, then the real axis maps to the real axis, and we thereby recover the ordinary exponential function. We also know that translating a horizontal line upwards will rotate its image ray counter-clockwise, but we don't yet know how fast. In [5.15] $d\theta$ is the infinitesimal rotation produced by moving z through a distance ϵ along the vertical edge of the square. But since the amplification is r , we know that the image of this edge has length $r\epsilon$, and consequently $d\theta = \epsilon$. In other words,

An infinitesimal vertical translation produces a numerically equal rotation. (5.23)

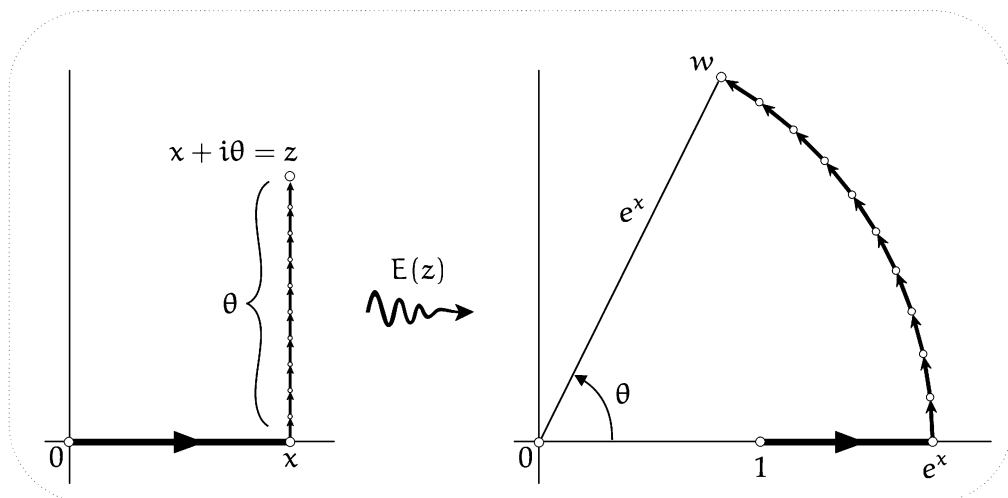
We can now completely describe the mapping produced by $E(z)$. Imagine watching the image as we move from the origin to a typical point $z = x + i\theta$ in a two-legged journey: first along the real axis to x , then straight up to z . See [5.16]. As we move to x , the image moves along the real axis from 1 to e^x . Repeated application of (5.23) then tells us that moving up a distance θ will rotate the image through an angle θ . For example, we find that $E(z)$ wraps the imaginary axis around the unit circle in such a way that

$$E(i\theta) = \cos \theta + i \sin \theta.$$

This is our old friend, the celebrated Euler Formula. It also follows directly from this geometry that the mapping has the property

$$E(a + b) = E(a) \cdot E(b).$$

It is now entirely logical to define " e^z " to be $E(z)$, and our work is done.



[5.16] Geometry of the exponential. If we insist that $E(0) = 1$, then combining the two facts we deduced in the previous figure, we see that we have arrived at the familiar geometry of $z \mapsto e^z$.

As we indicated at the start of this section, there is in fact an even more compelling explanation than the above. We have just used a very natural differential equation to propagate e^x off the real axis; however, it will turn out that even this equation is superfluous. The rigidity of analytic functions is so great that merely knowing the values of e^x on the real axis uniquely determines its “analytic continuation” into the complex realm.

5.9 An Application of Higher Derivatives: Curvature*

5.9.1 Introduction

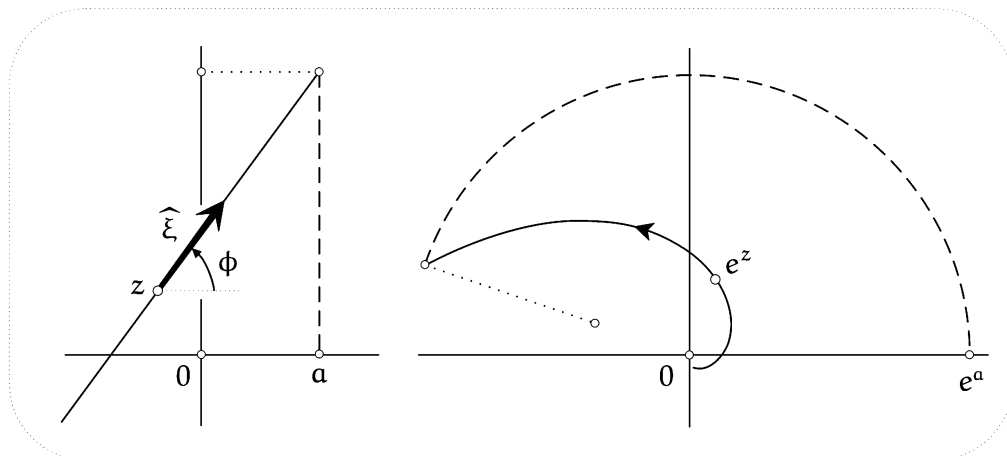
Earlier we alluded to the remarkable fact that analytic functions are infinitely differentiable. In other words, if f is analytic then f'' exists. In this section we seek to shed geometric light on the meaning and existence of this second derivative f'' . We shall do so by answering the following question:

If an analytic mapping f acts on a curve K of known curvature κ at p , then what is the curvature $\tilde{\kappa}$ of the image curve $\tilde{K}$ at $f(p)$?

In the next section we shall see that the solution to this problem provides a novel insight into (of all things!) the elliptical orbits of the planets round the sun at one focus.

At the risk of ruining the suspense, here is the answer to our question:

$$\tilde{\kappa} = \frac{1}{|f'(p)|} \left(\operatorname{Im} \left[\frac{f''(p) \hat{\xi}}{f'(p)} \right] + \kappa \right), \quad (5.24)$$



[5.17] Transformation of curvature under $z \mapsto e^z$. Applying the general formula (5.24) for the transformation of curvature, the image under $z \mapsto e^z$ of the straight line at angle ϕ is given by $\tilde{\kappa} = e^{-x} \sin \phi$. Check for yourself that this is in accord with the empirical evidence in this figure.

where $\hat{\xi}$ denotes the unit complex number tangent to the original curve at p . Before explaining this result, let us simply test it on an example.

On the left of [5.17] we have drawn three line-segments, and on the right their images under $f(z) = e^z$. The segments are distinguished by the value of the angle ϕ that each makes with the horizontal: $\phi = 0$ for the dotted one; $\phi = (\pi/2)$ for the dashed one; and the solid one represents a general value of ϕ . Now look at the curvature of their images: $\tilde{\kappa} = 0$ for the dotted one; $\tilde{\kappa} = e^{-a}$ for the dashed one; and on the solid image, $\tilde{\kappa}$ starts out large and then dies away as we spiral out from the origin.

In order to compare these empirical observations with our formula, write the unit tangent as $\hat{\xi} = e^{i\phi}$ and note that if $f(z) = e^z$ then $f'' = f' = e^z$. With $z = x + iy$, formula (5.24) therefore reduces to

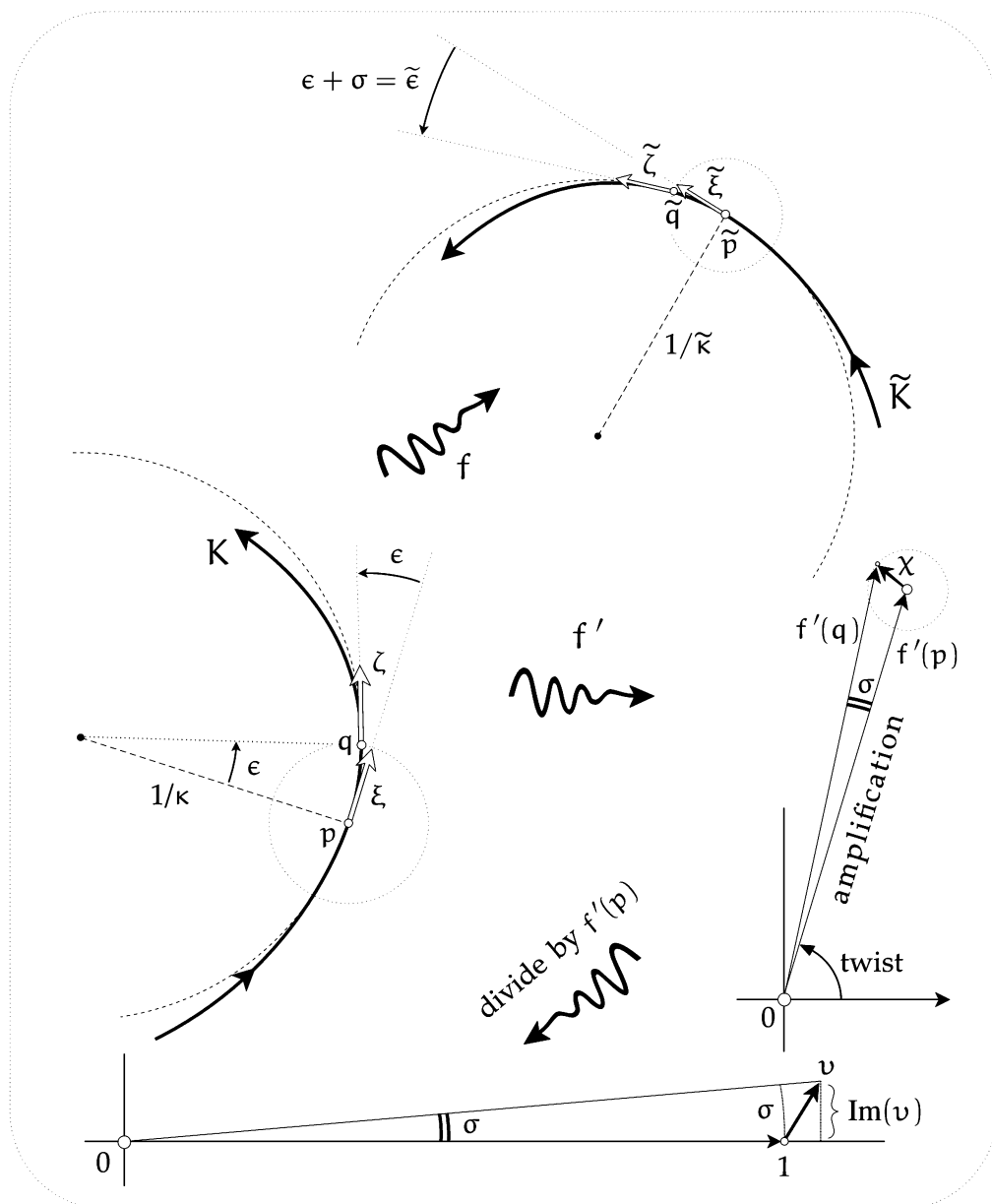
$$\tilde{\kappa} = e^{-x} (\sin \phi + \kappa).$$

Using the fact that $\kappa = 0$ for our line-segments, and that ϕ is constant on each, you may now easily check the accord between this formula and figure [5.17].

5.9.2 Analytic Transformation of Curvature

We now turn to the explanation of (5.24). The presence of an imaginary part in this rather daunting formula would seem to bode ill for a purely geometric attack. Surprisingly, this isn't the case. Consider [5.18].

On the left is the curve K , with curvature κ at p . Note that we have arbitrarily assigned a sense to K so as to give κ a definite sign. At the top of the figure is the



[5.18] Geometric derivation of the transformation law for curvature. The analytic mapping f transforms the curve K into the curve $\tilde{K}$. Curvature is the rate of rotation of the tangent, so $\kappa \asymp (\epsilon/|\xi|)$ at p is transformed into $\tilde{\kappa} \asymp (\tilde{\epsilon}/|\tilde{\xi}|) \asymp (\tilde{\epsilon}/|f'(p)\xi|)$ at $\tilde{p} = f(p)$. If the twists at p and q were the same, then the turning angle ϵ would not change, but in fact (as we see on the right) the amplitwists at p and q differ by $\chi \asymp f''(p)\xi$, so there is an extra twist $\sigma \asymp (\text{angle subtended by } \chi)$. To geometrically determine this extra twist σ , divide the figure on the right by $f'(p)$ to obtain the figure at the bottom. We now see that $\sigma \asymp \text{Im}(v) = \text{Im}\left[\frac{\chi}{f'(p)}\right] \asymp \text{Im}\left[\frac{f''(p)\xi}{f'(p)}\right]$, thereby completing the proof of the curvature transformation law, (5.24).

image curve $\tilde{K}$ under the mapping f ; note that its sense is determined by that of K . It is the curvature of $\tilde{K}$ at $\tilde{p} = f(p)$ that (5.24) purports to describe.

As illustrated, ξ is a small (ultimately infinitesimal) complex number tangent to K at p . With centre at p we have drawn a circle through the tip of ξ , cutting K in q . At q we have drawn another small (ultimately infinitesimal) tangent complex number ζ , and we have marked the angle ϵ of rotation from ξ to ζ . Recall that the curvature κ at p is, by definition, the rate of rotation of the tangent with respect to distance along K . Since for infinitesimal ξ , the arc pq equals $|\xi|$, the curvature at p is therefore

$$\kappa = \frac{\epsilon}{|\xi|}. \quad (5.25)$$

Likewise, at the image points $\tilde{p}$ and $\tilde{q}$ on the image curve $\tilde{K}$ we have drawn the image complex numbers $\tilde{\xi}$ and $\tilde{\zeta}$, the rotation from $\tilde{\xi}$ to $\tilde{\zeta}$ being $\tilde{\epsilon}$. Thus the image curvature is

$$\tilde{\kappa} = \frac{\tilde{\epsilon}}{|\tilde{\xi}|}. \quad (5.26)$$

Our problem therefore reduces to finding $\tilde{\epsilon}$ and $|\tilde{\xi}|$.

Since $|\tilde{\xi}|$ is the length of the amplitwisted image of ξ ,

$$|\tilde{\xi}| = (\text{amplification}) \cdot |\xi| = |f'(p)| \cdot |\xi|. \quad (5.27)$$

The more interesting and difficult part of the problem is to find $\tilde{\epsilon}$.

If ξ and ζ both underwent precisely the same twist, then the turning angle $\tilde{\epsilon}$ for the images would equal the original turning angle ϵ . However, the twist at q will differ very slightly, say by σ , from that at p . Thus

$$\tilde{\epsilon} = \epsilon + (\text{extra twist}) = \epsilon + \sigma. \quad (5.28)$$

This is how f'' enters the picture, for it describes how the amplitwist varies.

The function f' is a perfectly respectable mapping in its own right, and it may be drawn like any other. The right-hand side of [5.18] is precisely such a picture. Each point z is mapped to the complex number that amplitwists infinitesimal complex numbers emanating from z . In particular, we have drawn the images $f'(p)$ and $f'(q)$ of p and q . The statement about infinite differentiability can now be recast in a more blatantly astonishing form: *if f is locally an amplitwist, then f' automatically is too*. We have indicated this in the picture by showing the disc at p being mapped by f' to *another disc* at $f'(p)$. This startling fact will now yield to us the value of σ .

The amplitwist that carries the disc at p to the disc at $f'(p)$ is $f''(p)$. In particular, ξ is amplitwisted to

$$\chi = f''(p) \xi.$$

But looking at the triangle on the right, the sides of which are the known quantities $f'(p)$ and χ , we see that the angle at the origin is precisely the extra twist σ that we seek.

It is easier to obtain an expression for this angle if we first rotate the triangle to the real axis. This rotation is achieved quite naturally (see the bottom figure) by dividing by $f'(p)$; the sides of the triangle now become 1 and $v = [\chi/f'(p)]$. Because σ equals the (ultimately) vertical arc through 1, the figure tells us that

$$\sigma = \text{arc} = \text{Im}(v) = \text{Im} \left[\frac{\chi}{f'(p)} \right] = \text{Im} \left[\frac{f''(p) \xi}{f'(p)} \right].$$

Thus, from (5.26), (5.27), and (5.28), and taking evaluation at p as understood, we obtain

$$\tilde{\kappa} = \frac{\left(\text{Im} \left[\frac{f'' \xi}{f'} \right] + \epsilon \right)}{|f'| |\xi|}.$$

Finally, using (5.25) and noting that $\hat{\xi} = (\xi/|\xi|)$ is the unit tangent at p , we do indeed obtain formula (5.24).

Finally, we note that, alternatively, a very short (but unilluminating) computational proof can be obtained [exercise] by differentiation of the equation,

$$\text{Twist} = \text{Im} \log f'.$$

5.9.3 Complex Curvature

Let us take a closer look at formula (5.24), which may be written

$$\tilde{\kappa} = \text{Im} \left[\frac{f'' \hat{\xi}}{f' |f'|} \right] + \frac{\kappa}{|f'|}.$$

The presence of the second term can be understood as follows. If the plane were to undergo a uniform expansion by factor R then a circle of radius $(1/\kappa)$ would become a circle of radius (R/κ) , that is of curvature (κ/R) . But a small piece of a general curve resembles an arc of its *circle of curvature*⁴, and the principal local effect of f (apart from a curvature-preserving twist) is an expansion by factor $|f'|$.

In addition to this phenomenon, the first term says that the mapping will introduce curvature even when none is originally present: the curvature k of the image of a straight line (as a function of its direction) is

$$k(\hat{\xi}) = \text{Im} \left[\frac{f'' \hat{\xi}}{f' |f'|} \right].$$

Now consider the fate of all the curves that pass through p in the direction $\hat{\xi}$. The general formula says that f will not only scale their curvatures by $(1/|f'|)$ (as previously explained), but it will also increase their curvatures by the *fixed* amount $k(\hat{\xi})$. In this sense, the first term corresponds to an intrinsic property of the mapping f .

⁴ The circle that touches the curve at the point in question, and whose curvature $\kappa = (1/\text{radius})$ agrees with that of the curve at that point.

However, $k(\hat{\xi})$ is not really intrinsic to f since it retains a vestige of the original curves, namely, their direction $\hat{\xi}$. It would appear that the most natural intrinsic quantity that can be abstracted from $k(\hat{\xi})$ is

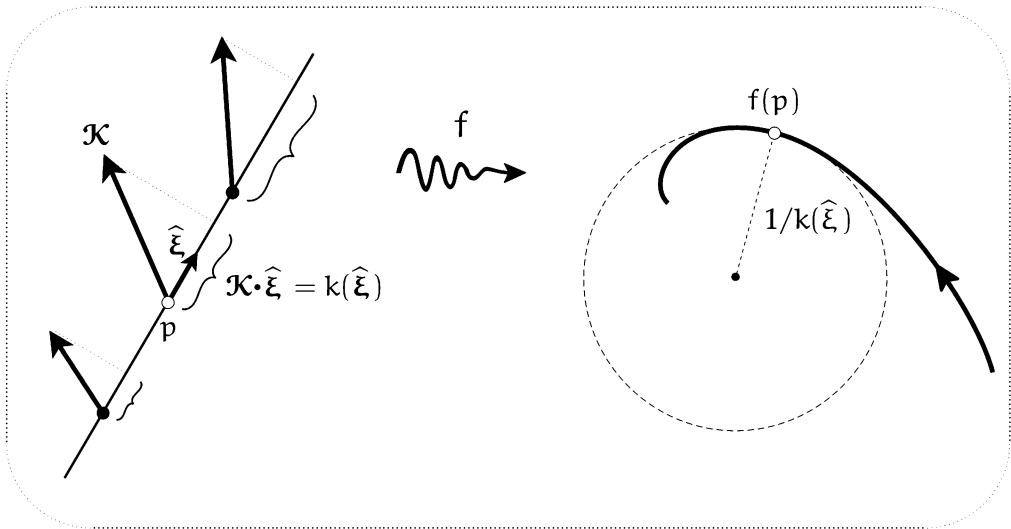
$$\mathcal{K} \equiv \frac{i \overline{f''}}{\overline{f'} |f'|}. \quad (5.29)$$

We propose to call this complex function $\mathcal{K}$ (which does not appear to have been investigated previously) the *complex curvature*⁵ of f .

To see that the complex curvature is indeed a natural quantity, picture $\mathcal{K}(p)$ as a vector emanating from p . We will show that

The projection of $\mathcal{K}(p)$ onto a line through p is the curvature of the image of that line at $f(p)$. (5.30)

See [5.19], in which $\mathcal{K}$ has also been drawn at two additional points. Note how the increasing length of the projection of $\mathcal{K}$ onto the line corresponds to increasing curvature along the image.



[5.19] Geometric interpretation of the complex curvature. *The projection of the complex curvature $\mathcal{K}(p)$ onto a line through p is the curvature of the image of that line at $f(p)$. Here, the projection grows as we move along the line, so the image curve bends more tightly.*

⁵ NOTE added in this 25th Anniversary Edition: I am delighted that my discovery of $\mathcal{K}$ and its mathematical properties has since found several applications (particularly in physics) that go beyond the two applications that I originally discovered and published here in VCA, namely, to dual central force fields [this chapter] and to the geometry of harmonic functions [Chapter 12]. For example, see Vitelli and Nelson (2006) and Mughal and Weaire (2009).

To prove (5.30), recall how the scalar product in $\mathbb{R}^2$ can be expressed in terms of complex multiplication:

$$\mathcal{K} \cdot \widehat{\xi} = \operatorname{Re} \left[\overline{\mathcal{K}} \widehat{\xi} \right] = \operatorname{Im} \left[i \overline{\mathcal{K}} \widehat{\xi} \right] = \operatorname{Im} \left[\frac{f'' \widehat{\xi}}{f' |f'|} \right] = k(\widehat{\xi}),$$

as was to be shown. This result yields a neater and more intelligible form of (5.24):

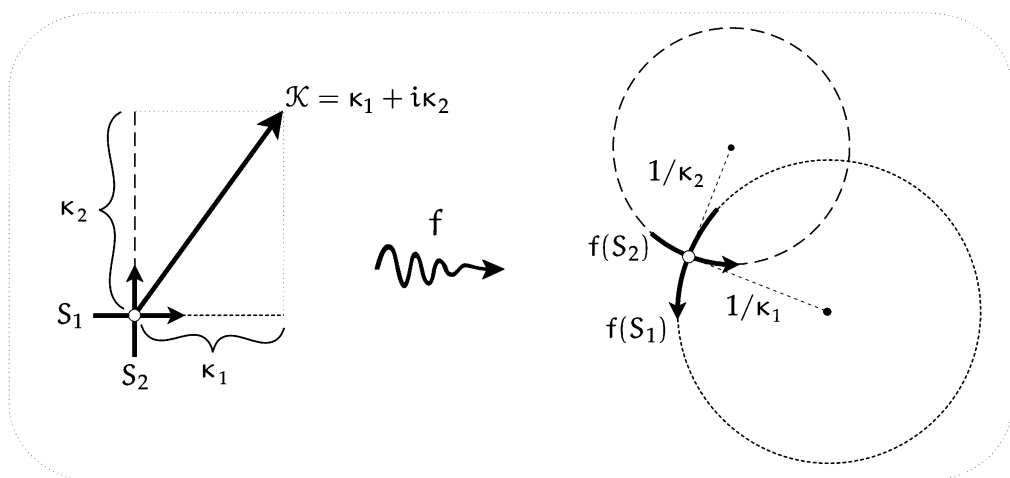
$$\tilde{\kappa} = \mathcal{K} \cdot \widehat{\xi} + \frac{\kappa}{|f'|}. \quad (5.31)$$

To see how $\mathcal{K}(p)$ may be determined geometrically, imagine a short, directed line-segment S rotating about p . The image $f(S)$ rotates with equal speed about $f(p)$, and its curvature oscillates sinusoidally: it reaches its maximum value $|\mathcal{K}(p)|$ when S points in the direction of $\mathcal{K}(p)$, while it vanishes when S is perpendicular to $\mathcal{K}(p)$.

In fact, to reconstruct $\mathcal{K}(p)$ it is sufficient to know the image curvatures κ_1 and κ_2 for just two positions S_1 and S_2 of the line-segment. Figure [5.20] illustrates this in the particularly simple case that S_1 and S_2 are horizontal and vertical, respectively. We then have

$$\mathcal{K} = \kappa_1 + i\kappa_2.$$

We conclude this section with a different way of looking at $\mathcal{K}$. On the left of [5.21] is an infinitesimal black shape Q , together with copies obtained by translating Q a fixed amount $|\xi|$ in various directions ξ . Under an analytic mapping f , Q is amplitwisted to the similar black shape $\tilde{Q}$ on the right. As Q translates by ξ , $\tilde{Q}$ not only translates by $f'\xi$, but it also rotates and expands. More precisely, the rotation



[5.20] Geometric interpretation of the real and imaginary parts of the complex curvature. The real and imaginary parts of the complex curvature $\mathcal{K}(p)$ are the curvatures of the images of the real and imaginary line segments through p .

of $\tilde{Q}$ is just the angle σ on the RHS of [5.18]. This rotation is clearly greatest when χ is perpendicular to $f'(p)$, pointing counterclockwise along the circle $|f'| = \text{const.}$ This occurs when ξ is in the direction of $\mathcal{K}$, for then

$$\chi \propto f'' \mathcal{K} \propto i/\bar{f}' \propto if'.$$

If we turn the direction of motion of Q by $-(\pi/2)$, then χ also turns by $-(\pi/2)$ to point radially outwards along the ray $\arg f' = \text{const.}$, thereby producing the greatest increase in $|f'|$.

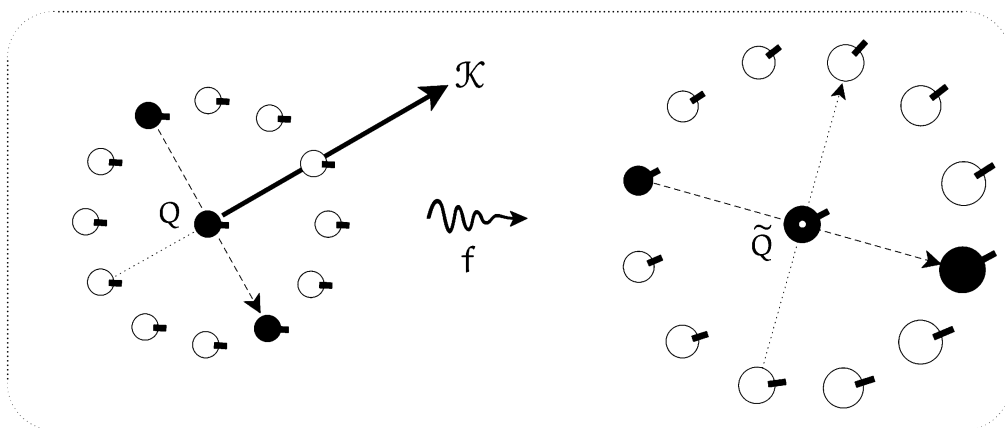
We now understand [5.21] in greater detail:

Let Q be an infinitesimal shape, and let $\tilde{Q}$ be its image under an analytic mapping f . Then $\tilde{Q}$ rotates most rapidly, and its size remains constant, when Q moves in the direction of $\mathcal{K}$. On the other hand, $\tilde{Q}$ expands most rapidly, and does not rotate, when Q moves in the orthogonal direction $-i\mathcal{K}$. (5.32)

In still greater detail, as Q begins to translate in an arbitrary direction $\hat{\xi}$, let $\mathcal{R}$ denote the rate of rotation of $\tilde{Q}$ with respect to the distance it moves. Then

$$\mathcal{R} = \mathcal{K} \cdot \hat{\xi}.$$

This achieves its maximum value $\mathcal{R}_{\max} = |\mathcal{K}|$ when Q moves in the direction of $\mathcal{K}$. Similarly, consider the expansion of $\tilde{Q}$. Let $\mathcal{E}$ denote the rate of increase of the size⁶



[5.21] Another geometric interpretation of the complex curvature. *Let Q be an infinitesimal shape, and let $\tilde{Q}$ be its image under an analytic mapping f . Then $\tilde{Q}$ rotates most rapidly, and its size remains constant, when Q moves in the direction of the complex curvature $\mathcal{K}$. On the other hand, $\tilde{Q}$ expands most rapidly, and does not rotate, when Q moves in the orthogonal direction $-i\mathcal{K}$.*

⁶ Here we mean the linear dimensions of $\tilde{Q}$. For example, if $\tilde{Q}$ were a disc then we could take its “size” to be its radius.

of $\tilde{Q}$ (again with respect to the distance *it* moves) as a fraction of $\tilde{Q}$'s initial size. Then [exercise]

$$\mathcal{E} = \widehat{\xi} \times \mathcal{K}.$$

This achieves its maximum value $\mathcal{E}_{\max} = |\mathcal{K}|$ when Q moves in the direction $-i\mathcal{K}$. These two results may be viewed as two facets of the single complex equation

$$\overline{\widehat{\xi}} \mathcal{K} = \mathcal{R} + i\mathcal{E}.$$

In Chapter 12, having developed the physical concepts of “flux” and “circulation”, we shall return to the complex curvature and see that it has other elegant properties and applications.

5.10 Celestial Mechanics*

5.10.1 Central Force Fields

If a particle p , moving through space, is constantly being pulled towards (or pushed away from) a fixed point o with a force that depends only on its distance r from o then we say that it is in a *central force field* and that o is the *centre of force*. No matter how the force varies with r , it is not hard to show [exercise] that the orbit of p will always lie in a plane through o .

Another feature of motion in *any* central force field is that the radius op sweeps out area at a constant rate $\mathcal{A}$, called the *areal speed*. A proof of this is given in Ex. 24. If the mass of p is m then [exercise] the angular momentum h of p is $2m\mathcal{A}$. The fact that $\mathcal{A}$ is constant is thus a manifestation of the conservation of angular momentum.

In addition to the angular momentum, the total energy E of the particle remains constant as it orbits. Henceforth, we shall always use a particle of unit mass. Thus if the particle's speed is v then the kinetic energy contribution has the definite value $\frac{1}{2}v^2$, while the potential energy contribution is only defined up to a constant. We shall restrict ourselves to force fields that vary as a power of r , and we may then fix the constant by arbitrarily assigning zero potential energy to the point where the field vanishes: if the force grows as a positive power of r , at the origin; if the force dies away as a negative power of r , at infinity.

5.10.2 Two Kinds of Elliptical Orbit

Consider the attractive *linear* force field in which, by definition, the force towards o is proportional to r . This linear force law is extremely important in physics, for if almost any physical system is slightly disturbed from equilibrium then the restoring force is precisely of this kind. Here is a simple example of what we mean; it will enable you to experimentally investigate motion in a linear force field. You are encouraged to do the following, not merely to imagine it.

Take a small weight W and suspend it just above a point o of a horizontal table using several feet of thread, perhaps attached to the ceiling. If you pull W to the side by just an inch or two then, because the thread is long, W barely rises above the table's surface and we may idealize this to a movement *on* the table. Furthermore, although the forces acting on W in this displaced position are actually gravity and the tension in the thread, the net effect [exercise] is as though o were magically pulling W towards it with a force proportional to r , as was required. To avoid the possibility of confusion later, we stress that gravity is playing absolutely no essential role here; it is merely providing one particularly convenient way of simulating a linear force field.

Now pull W a little bit away from o and give it a gentle flick in a random direction. You see that the orbit of W is a closed curve traversed again and again—a beautifully symmetrical oval shape centred at o . But exactly what is this oval?

It is an ellipse! To demonstrate this, take the tabletop to be $\mathbb{C}$ with o as its origin. Once again take W to have unit mass, and let its location at time t be $z(t)$. For simplicity's sake, let the force directed towards the origin *equal* the distance $|z|$. The differential equation governing the motion of W will therefore be $\ddot{z} = -z$, the two basic solutions of which are $z = e^{\pm it}$. These represent counter-rotating motions of unit speed around the unit circle. [Try launching W so as to produce these solutions.] The general solution is then obtained as a linear combination of these motions:

$$z = p e^{it} + q e^{-it}, \quad (5.33)$$

where p and q may, without any real loss of generality, be taken as real and satisfying $p > q$.

As is illustrated in [5.22], the addition of such counter-rotating circular motions results in elliptical motion with the attracting point at the centre. This becomes clear if we rewrite (5.33) as

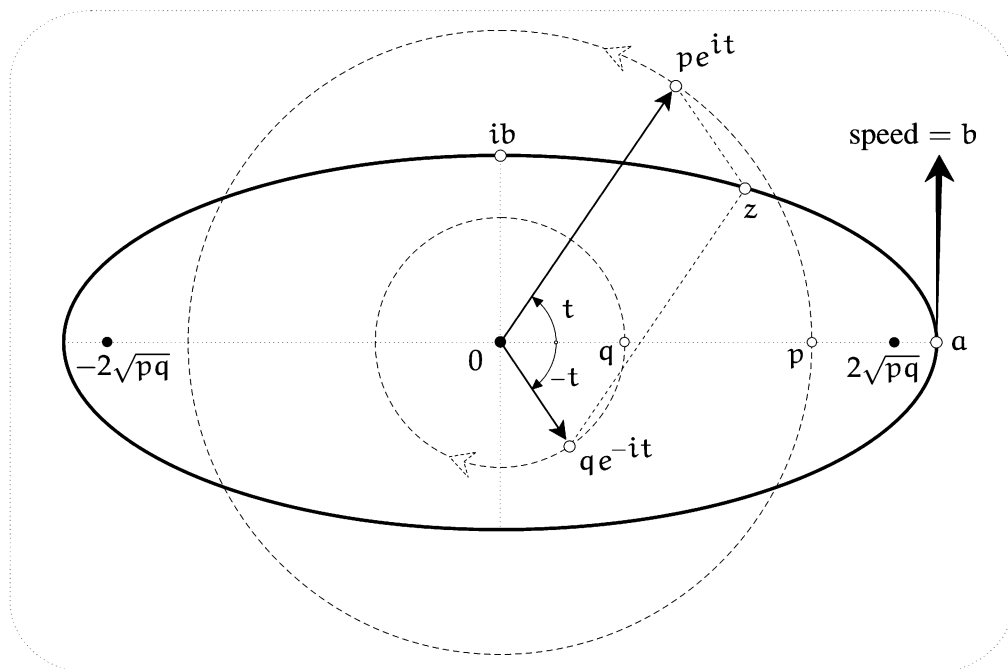
$$z = a \cos t + ib \sin t,$$

where $a = p + q$ and $b = p - q$. Each of these numbers has a double significance: a is both the semimajor axis and the point of launch; b is both the semiminor axis and the speed of launch. Note that the foci are at $\pm\sqrt{a^2 - b^2} = \pm 2\sqrt{pq}$.

Finally, for future use, let us calculate the constant energy E of a particle orbiting in this field. The potential energy is the work needed to pull the particle away from the origin out to a distance of r , namely, [exercise] $(r^2/2)$. Thus

$$E = \frac{1}{2}(v^2 + r^2).$$

As the particle orbits round the ellipse in [5.22], we see that this expression always equals $\frac{1}{2}(a^2 + b^2)$.



[5.22] Addition of counter-rotating circular motions results in elliptical motion.

We now turn to a second, more famous example of elliptical motion in a central force field: the orbits of the planets around the sun. There are two fundamental differences between this phenomenon and the one above. First, instead of the force of attraction increasing linearly with distance, here the force of gravity dies away as the square of the distance from the sun. Second, instead of the centre of attraction being at the centre of the elliptical orbit, here the sun is at one of the foci.

The ancient Greeks discovered that the ellipse has beautiful *mathematical* properties; two thousand years later Newton revealed that it has equally beautiful *physical* significance. He discovered that if, and *only*⁷ if, the force field is linear or inverse-square, then elliptical orbits result. In the *Principia* Newton explicitly drew attention to this coincidence, calling it “very remarkable”. As the Nobel physicist Subrahmanyan [Chandrasekhar](#) (1995, p. 287) observed, “nowhere else in the *Principia* has Newton allowed himself a similar expression of surprise.”

We are left with something of a mystery. There appears to be some special connection between the linear and inverse-square force fields, but what could it possibly be? Newton himself was able to find a connection, and we shall use complex analysis to find another. For more on both these connections, see [Arnol'd](#) (1990), [Needham](#) (1993), and [Chandrasekhar](#) (1995).

⁷ Newton assumed that the force varies as a power of the distance, but it has since been discovered that the result is still true if we drop this requirement.

5.10.3 Changing the First into the Second

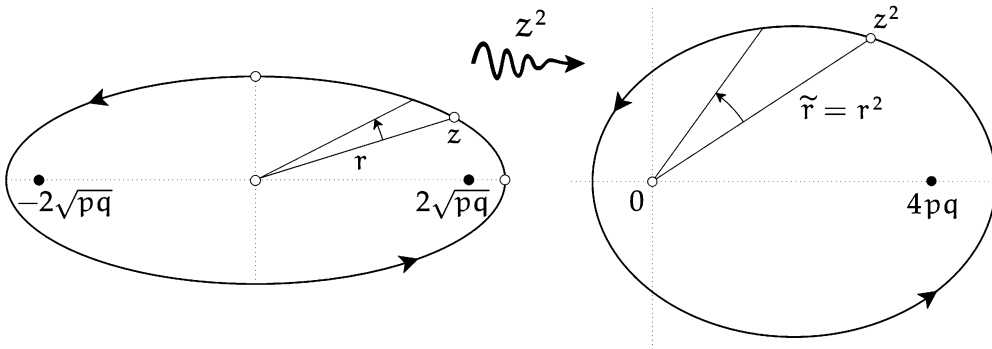
The geometry of complex numbers was not yet understood in the time of Newton; had it been, he would surely have discovered the following surprising fact. If we apply the mapping $z \mapsto z^2$ to an origin-centred ellipse, then the image is not some strange ugly shape, as one might expect, but rather another *perfect ellipse*; furthermore, this ellipse automatically has one focus at the *origin*! See [5.23]. Before exploring the implications, let us verify this fact: squaring (5.33),

$$z \mapsto z^2 = (p e^{it} + q e^{-it})^2 = p^2 e^{i2t} + q^2 e^{-i2t} + 2pq.$$

The first two terms correspond to an origin-centred ellipse with foci at $\pm 2pq$; the last term therefore translates the left-hand focus to the origin.

Expressed in dynamical terms, this geometric result states that while leaving the attracting point fixed at the origin, $z \mapsto z^2$ transforms an orbit of the linear field into an orbit of the inverse square field. However, we are only in a position to state the result in this way because we already know what the orbits in the two fields look like. *Is there instead some a priori reason why $z \mapsto z^2$ should map orbits of the linear field to orbits of the gravitational field?* If there were such a reason then [5.23] could be viewed as a novel derivation, or explanation, of the elliptical motion of planets about the sun as focus.

That there is indeed such a reason was discovered in the first decade of the 20th century. Several people deserve credit for this beautiful result which, at the time of this writing, is still not widely known. Apparently Bohlin (1911) was the first to publish it, not knowing that Kasner (1913) had already discovered a more general result in 1909. Finally, knowing only of Bohlin's work, Arnol'd (1990) rediscovered Kasner's general theorem.



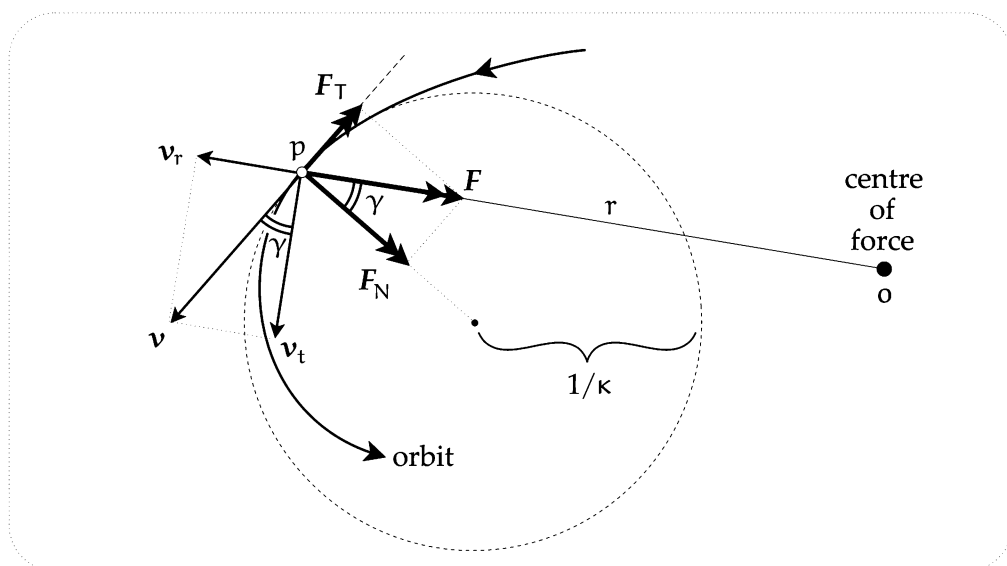
[5.23] Transformation of the linear force field into the inverse-square force field. *Very remarkably, squaring an origin-centred ellipse results in another ellipse. Furthermore, one of the foci of the image ellipse is located at the origin! As the text explains, this provides a novel explanation of the elliptical orbits of the planets around the Sun at one focus.*

Before embarking on the details of the explanation, here (following [Needham \(1993\)](#)) is our plan of attack. In the absence of force a particle will move in a straight line; *bending* is therefore the manifestation of force, and this can be quantified in terms of the curvature of the orbit. Since the mapping $z \mapsto z^2$ is analytic, we may use the results of the previous section to find the relationship between the curvature of an orbit and the curvature of the image orbit produced by the mapping. This will enable us to find the relationship between the forces that hold the preimage and image in their respective orbits.

5.10.4 The Geometry of Force

Given an orbit and a centre of force, our aim is to find a purely geometric formula for the magnitude F of the force F that holds the particle in that orbit. Consider figure [5.24]. As illustrated, it is conceptually very helpful to decompose F into components F_T and F_N that are tangential and normal to the orbit, respectively. The effect of the component F_T is to change the speed v of p without altering its course. The effect of the component F_N is to bend the orbit of p without altering its speed.

From elementary mechanics we know that if a particle of unit mass moves at constant speed v round a circle of radius ρ then the force directed towards the centre is (v^2/ρ) . Thus if the curvature of the orbit is κ (as illustrated) then $F_N = \kappa v^2$. If we call the acute angle between the radius and the normal γ , then it follows that the



[5.24] Geometric determination of the central force law from the orbit. If h is the (constant) angular momentum of the orbiting particle, the magnitude of the central force field that holds the particle in its orbit of curvature κ is given by $F = h^2 \left(\frac{\kappa \sec^3 \gamma}{r^2} \right)$.

total force acting on p is

$$F = F_N \sec \gamma = \kappa v^2 \sec \gamma.$$

In order to fully reduce this formula to geometric terms, we need to express v in geometric terms. This is made possible by the constancy of the angular momentum $h = 2A$. If we decompose the velocity v into radial and transverse components v_r and v_t , then clearly only the latter generates area. More precisely, $h = 2A = r v_t = r v \cos \gamma$, so

$$v = h \left(\frac{\sec \gamma}{r} \right). \quad (5.34)$$

Substituting for v in the previous result, we obtain the desired geometric formula for the force:

$$F = h^2 \left(\frac{\kappa \sec^3 \gamma}{r^2} \right). \quad (5.35)$$

This result is essentially due to [Newton \(1687, Prop. VII\)](#). Observe that the concept of time has almost disappeared in this formula, the only vestige being the constant h that specifies how fast the orbit is traversed.

5.10.5 An Explanation

As z describes an arbitrary orbit, (5.35) tells us the force F needed to hold it in that orbit. Now apply the mapping $z \mapsto z^2$, and let a tilde denote a quantity associated with the image, e.g., $\tilde{r} = r^2$. The force $\tilde{F}$ needed to hold the image in its orbit is

$$\tilde{F} = \tilde{h}^2 \left(\frac{\tilde{\kappa} \sec^3 \tilde{\gamma}}{\tilde{r}^2} \right),$$

and we now seek to relate this to the original force F .

First, to find $\tilde{\kappa}$, simply put $f(z) = z^2$ into (5.24) and thereby obtain [exercise]

$$\tilde{\kappa} = \frac{1}{2} \left[\frac{\cos \gamma}{r^2} + \frac{\kappa}{r} \right].$$

Next, observe that since the ray from 0 to z maps to the ray from 0 to z^2 , the conformality of the mapping implies that $\tilde{\gamma} = \gamma$.

Putting these facts into the formula for $\tilde{F}$, and substituting for the original speed and force from (5.34) and (5.35), we get

$$\tilde{F} = \left(\frac{\tilde{h}}{h} \right)^2 \frac{[\frac{1}{2}v^2 + \frac{1}{2}rF]}{\tilde{r}^2}. \quad (5.36)$$

Even if F is a simple power law, generally this $\tilde{F}$ will not be. However, if and *only* if the original force field is *linear*⁸, the numerator in the above expression magically

⁸ However, as we shall see in a moment, it could be the *repulsive* linear field $F = -r$ instead of the attractive one $F = +r$.

becomes the constant total energy E of the particle in the original field:

$$\tilde{F} = \left(\frac{\tilde{h}}{h} \right)^2 \frac{E}{\tilde{r}^2} ! \quad (5.37)$$

The image therefore moves in a field that is *inverse-square*, as was to be shown.

Here is a fact which may have been bothering you already. The only gravitational orbits we have managed to explain in this way are the ellipses; where are the hyperbolic orbits which we know are also possible in a gravitational field? In fact the geometry of $z \mapsto z^2$ *does* explain these, the resolution being that gravitational orbits arise not only as the images of orbits in a linear field that is attractive, but also of orbits in a linear field that is *repulsive*, $F = -r$. The orbits in this field are hyperbolae with centre (i.e., intersection of asymptotes) at the origin, and $z \mapsto z^2$ maps these to hyperbolae with one focus at the origin.

The dynamical explanation is almost unchanged: the constant total energy of the particle in the original repulsive linear field is now given by $E = \frac{1}{2}(v^2 - r^2)$, so inserting $F = -r$ in (5.36) once again yields (5.37). See [Needham \(1993\)](#) for more on this, as well as the general result we are about to state, which may be proved in exactly the same way as the special case above.

5.10.6 The Kasner–Arnol'd Theorem

The power laws $F \propto r$ and $\tilde{F} \propto \tilde{r}^{-2}$ are examples of what Arnol'd calls *dual* force laws, and both he and Kasner discovered that they constitute just one example of duality. Here is the general result:

Associated with each power law $F \propto r^A$ there is precisely one power law $\tilde{F} \propto \tilde{r}^{\tilde{A}}$ that is dual in the sense that orbits of the former are mapped to orbits of the latter by $z \mapsto z^m$, and the relationships between the forces and the mapping are:

$$(A + 3)(\tilde{A} + 3) = 4 \quad \text{and} \quad m = \frac{(A + 3)}{2}.$$

To their result we add the following point of clarification on the role of energy:

In general, positive energy orbits in either the attractive or repulsive field $F \propto r^A$ map to attractive orbits in the dual field, while negative energy orbits map to repulsive ones. However, if $-3 < A < -1$ (e.g., gravity) then these roles are reversed. In all cases, zero energy orbits map to force-free rectilinear orbits.

5.11 Analytic Continuation*

5.11.1 Introduction

Throughout this book we have stressed how functions may be viewed as geometric entities that need not be expressed (nor even be expressible) in terms of formulae. As an illustration of the limitations of formulae, consider

$$G(z) = 1 + z + z^2 + z^3 + \dots$$

This power series converges inside the unit circle $|z| = 1$, and consequently it is analytic there. Figure [5.25] shows a grid of little squares inside this circle being amplitwisted by $z \mapsto w = G(z)$ to another such grid lying to the right of the vertical line $\operatorname{Re}(w) = \frac{1}{2}$, which itself is the image of the circle. Now this circle is certainly a barrier to the power series formula for G , since it clearly diverges at 1: geometrically, the image of the circle extends to ∞ . However, the circle is *not* a barrier to the geometric entity that the formula is attempting to describe.

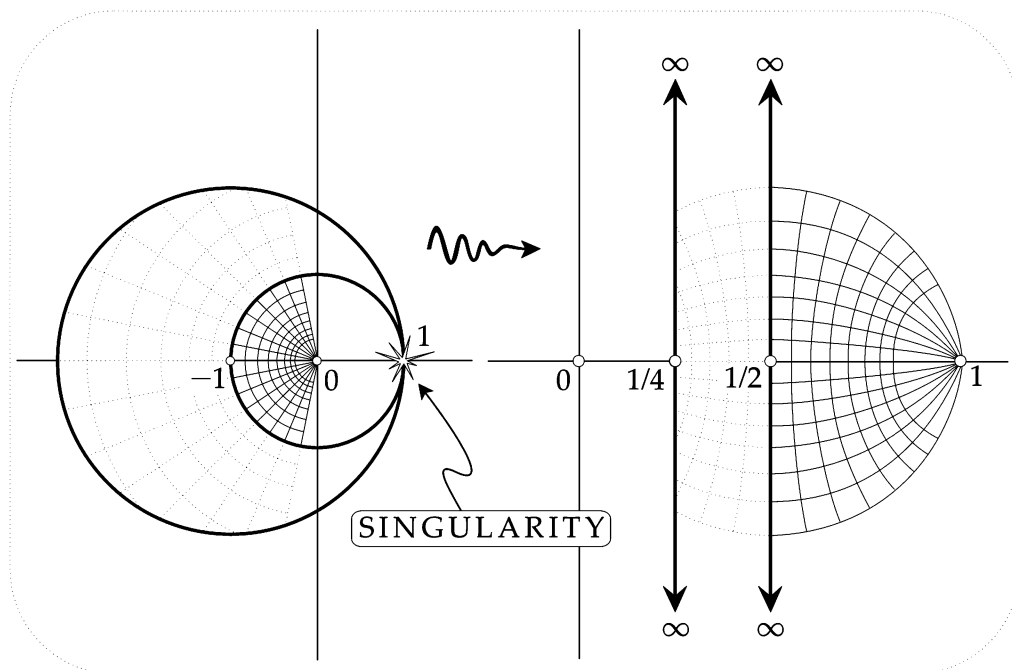
Consider a somewhat different-looking power series centred at -1 :

$$H(z) = \frac{1}{2} \left[1 + \left(\frac{z+1}{2} \right) + \left(\frac{z+1}{2} \right)^2 + \dots \right].$$

This series is analytic inside a larger circle of convergence $|z+1| = 2$. Despite the apparent difference, $H(z)$ maps the previously considered solid grid inside $|z| = 1$ to precisely the same grid on the right of $x = (1/2)$ as G did: $H = G$ inside $|z| = 1$. But now the grid may be extended to the dotted one lying *outside* $|z| = 1$, and H amplitwists it to the dotted grid lying to the left of $x = (1/2)$. We say that H is an *analytic continuation* of G to the larger disc. An obvious question is whether H is the *only* analytic continuation of G to this region. As we hope [5.25] makes palpable, the rigidity imposed by being locally an amplitwist does indeed force the mapping to grow in a unique way.

The objective of this final section will be to make this rigidity clearer, and also to describe one method (due to Schwarz) of explicitly finding the mapping in regions beyond its original definition. Before doing this, however, we will complete our discussion of [5.25].

The figure makes it plain that H is no more the end of the line than G was: it too can be continued. But if we cling to power series then the scope of our description of the mapping that underlies both G and H will be strictly limited. This is because such series only converge inside discs, and if we try to expand any disc then it will eventually hit the singularity at $z = 1$ and then be unable to go round it. Thus any power series will necessarily miss out at least half of the potential domain of the mapping. On the other hand, as you may have already noticed, the Möbius transformation $1/(1-z)$ is analytic everywhere except $z = 1$, and it agrees with both



[5.25] Analytic Continuation. The origin-centred power series $G(z) = 1 + z + z^2 + z^3 + \dots$ only converges inside the unit disc. The image $w = G(z)$ of the grid of small “squares” on the left (inside the unit disc) is shown [solid] on the right, in the region $\operatorname{Re}(w) > \frac{1}{2}$. But if we extend the grid [dotted] on the left beyond the unit circle, then there is a unique analytic continuation of the image grid [dotted] on the right into the region $\frac{1}{4} < \operatorname{Re}(w) < \frac{1}{2}$, expressible as a power series that converges inside the larger circle $|z + 1| = 2$. If we abandon power series altogether, then the grid can in fact be extended uniquely to the entire plane, the complete analytic continuation being given by $z \mapsto 1/(1 - z)$, with the singularity $z = 1$ being mapped to the point at ∞ .

G and H within their circles of convergence; it thus constitutes the complete analytic continuation of the mapping. [We encourage you to use this fact to check the details of the figure.] The simplicity of this example is perhaps misleading. Usually one cannot hope to capture the entire geometric mapping within a single closed expression such as $1/(1 - z)$.

When one stares at a figure like [5.25] one starts to sense the rigid growth of the mapping due to the analytic requirement that an expanding mesh of tiny squares must map to another such mesh. It also becomes clear how the mapping itself is oblivious to the different formulae with which we try to describe it. Indeed we have seen that the two circles—such formidable and impenetrable barriers to the power series—have only a slight significance for the mapping itself: both hit $z = 1$, so both images extend to ∞ .

5.11.2 Rigidity

The essential character of analytic rigidity is captured in the following result:

*If even an arbitrarily small segment of curve is crushed to a point by an analytic mapping, then its **entire domain** will be collapsed down to that point.*

The theory of integration to be developed in the following chapters will provide a convincing explanation of this fact. For the present, though, we can obtain a good measure of insight into its truth by extending our previous discussion of critical points (page 231). This may give the illusion of dispensing with integration theory, but as we pointed out at the time, that discussion *also* had to draw on future results. We now recap the relevant facts concerning critical points.

The amplification vanishes at a critical point p , leading to the impression that an infinitesimal disc centred there is crushed down to a point. However, this is merely a ‘trick of the light’ due to low magnification of the image plane. If the order of p is $(m-1)$, so that the mapping locally resembles z^m , then an infinitesimal disc at p of radius ϵ will be mapped [m -fold] onto a vastly smaller disc of radius ϵ^m . In terms of the microscope analogy this means that we must use the L_m lens to see that the image isn’t a perfect point. The greater the order, the greater the degree of crushing at p , and the greater the power of the first lens that will reveal the nonpointlike image.

Now observe that, calculationally speaking, the role of the increasingly high-powered lenses that fail to resolve the image is taken over by the increasingly high-order derivatives that vanish at p :

$$f(z) \sim z^m \iff \begin{cases} L_1, \dots, L_{m-1} \text{ show nothing, but image visible with } L_m \\ f'(p) = 0, f''(p) = 0, \dots, f^{(m-1)}(p) = 0, \text{ but } f^{(m)}(p) \neq 0 \end{cases}$$

In short, the higher the derivative that vanishes at p , the greater the degree of crushing at p .

We now apply this insight to the given situation. Let s be the (possibly) tiny segment that is crushed by $f(z)$. The amplification of f at a point of s may be read off by looking in any direction. By choosing to look *along* s we find that the amplification vanishes at each point of s . The entire segment is therefore made up of critical points for which $f' = 0$. Now think of f' as an analytic mapping in its own right, just as we did in [5.18]. We have just seen that this mapping automatically possesses the same property as f did: it crushes s to a point. We conclude that *its* derivative must also vanish on s . Clearly there is no end to this; *all* the derivatives of f must vanish, and, correspondingly, infinitesimal discs centred on s must be *totally* crushed.

This means that there is at least a sheathlike region surrounding s which is completely crushed by f . But if we take a new curve lying in this region, the whole line

of thought may be repeated to deduce that f must crush a still larger region. The collapse of the function therefore proceeds outwards (at the speed of thought!) to the entire domain.

5.11.3 Uniqueness

Suppose that $A(z)$ and $B(z)$ are both analytic functions defined on a region that happens to be the same size and shape as California. Suppose, further, that A and B both happen to have the same effect on a tiny piece of curve, say a fallen eyelash lying in a San Francisco street. This tiny measure of agreement instantly forces them into *total* agreement, even hundreds of miles away in Los Angeles! For $(A - B)$ is analytic throughout California, and since it crushes the eyelash to 0, it must do the same to the entire state.

We can express this slightly differently. If we arbitrarily specify the image points of a small piece of curve s , then in general there will not exist an analytic function that sends s to this image. However, the previous paragraph assures us that if we can find such a function on a domain including s , then it is *unique*.

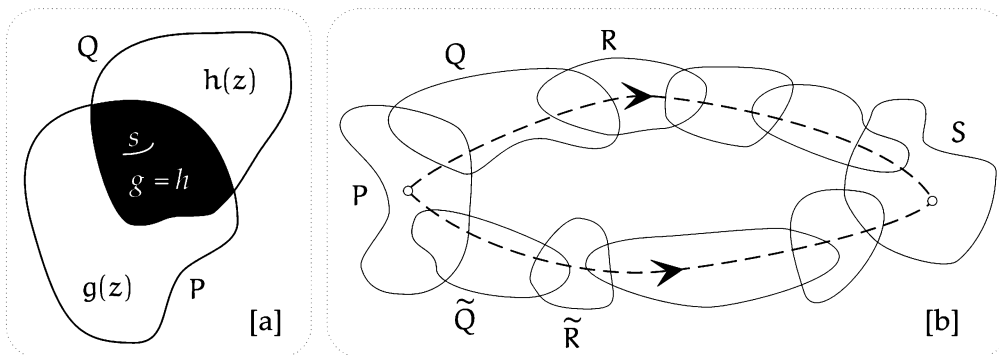
This is the “compelling reason” we referred to earlier in connection with the uniqueness of the generalization of e^x to complex values. For if an analytic generalization $E(z)$ exists, then we see that it will be uniquely determined by the values of e^x on even a small piece of the real axis. Of course knowing this does not help in the least to find out what $E(z)$ actually *is*. The value of our previous derivations of explicit expressions for $E(z)$ therefore remains undiminished. On the other hand, the new knowledge is not without practical implications. Consider these three very different-looking expressions:

$$\lim_{n \rightarrow \infty} \left(1 + \frac{z}{n}\right)^n, \quad e^x (\cos y + i \sin y), \quad 1 + z + z^2/2! + z^3/3! + \dots$$

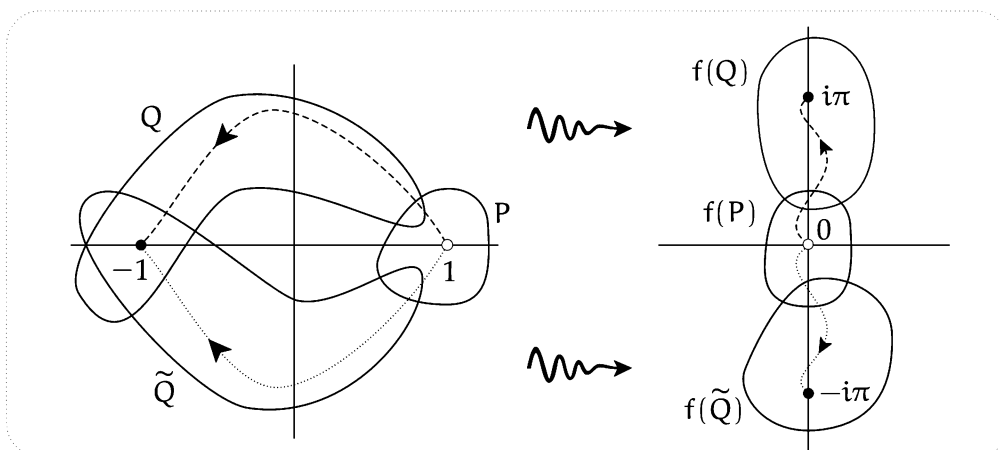
They are all analytic, and they all agree with e^x when z is real. Thus, without further calculation, we know they must all be equal to each other, for they can only be different ways of expressing the unique analytic continuation of e^x .

New and important aspects of uniqueness emerge when we consider domains that merely overlap, rather than coincide. Let $g(z)$ and $h(z)$ be analytic functions defined on the sets P and Q shown in [5.26a]. If they agree on even a small segment s in $P \cap Q$ then they will agree throughout $P \cap Q$. If we imagine that we initially only know about g on P , then we may think of h as describing the same geometric mapping as g but with the domain P extended to encompass Q . We are encouraged in this view by the fact that g uniquely determines this analytic continuation. For suppose h^* were another continuation of g into Q . On s we would then have $h^* = g = h$, but this forces $h^* = h$ throughout their common domain Q .

The functions $G(z)$ and $H(z)$ of the introduction furnish a concrete example of the above, where P happens to lie wholly within Q . The function $1/(1 - z)$ then constitutes the analytic continuation of H to the rest of the plane.



[5.26] Analytic continuation along alternative routes. [a] If $g(z)$ and $h(z)$ are analytic functions defined in overlapping regions P and Q , and they agree on even a small segment s in this overlap, then they must agree throughout $P \cap Q$, and we are entitled to think of $h(z)$ as the unique analytic continuation of $g(z)$ from P into Q . [b] Although the analytic continuations along $PQR \cdots S$ and along $P\tilde{Q}\tilde{R} \cdots S$ are both uniquely determined, they need not agree on S : in this case we have a multfunction.



[5.27] Analytic continuation of $\log z$. In P we can define a single-valued branch of $\log z$ given by $f(z) = \ln r + i\theta$, where $-\pi < \theta \leq \pi$. The figure shows how 1 maps to 0, and how P maps to the surrounding region $f(P)$. The definition of $\log z$ then has unique analytic continuations into Q and into $\tilde{Q}$, but they disagree at -1 , for example.

Just as g was continued from P to Q , so we may continue the process along a whole chain of overlapping sets $P, Q, R, \dots, S$, as in [5.26b]. We thereby obtain a unique analytic continuation of g to S . But what if we chose an alternative route such as $P, \tilde{Q}, \tilde{R}, \dots, S$? Once again the continuation of g to S is unique, *but this need not agree with the first continuation*. The idea of analytic continuation has thus led very naturally to the idea of multivalued functions.

Consider [5.27]. In P we can define a single-valued branch of $\log z$ given by $f(z) = \ln r + i\theta$, where $-\pi < \theta \leq \pi$. The figure shows how 1 maps to 0, and how P maps to the surrounding region $f(P)$. If we define $g(z) = \ln r + i\Theta$ (with $-\frac{\pi}{2} < \Theta \leq \frac{3\pi}{2}$)

then since $g = f$ on $P \cap Q$, it must be the analytic continuation of f to Q . Likewise its continuation to $\tilde{Q}$ is $\tilde{g}(z) = \ln r + i\tilde{\Theta}$, where $-\frac{3\pi}{2} < \tilde{\Theta} \leq \frac{\pi}{2}$, for example. In the region surrounding -1 we now have two unique continuations of one and the same function f . But despite this common ancestry, they clearly disagree with each other: $g(-1) = i\pi$, while $\tilde{g}(-1) = -i\pi$.

5.11.4 Preservation of Identities

In this subsection we will show that any identities that hold for real functions must continue to hold for their analytic generalizations to $\mathbb{C}$ (assuming such exist). This is easiest to explain through examples.

First we consider an important example dealing with power series. Suppose that the real function $f(x)$ can be represented by a convergent power series

$$f(x) = a + bx + cx^2 + dx^3 + \cdots.$$

We therefore know that the complex series

$$F(z) = a + bz + cz^2 + dz^3 + \cdots$$

is convergent and hence analytic. But since $F(x) = f(x)$ on the real axis, it follows that F is the unique analytic continuation of f to complex values. In other words, the transition from f to its analytic continuation does not change the formula (series).

For our next example we consider a real identity involving *two* variables: $e^x \cdot e^y = e^{x+y}$. It will help to appreciate the argument if you can be temporarily stricken with amnesia, so that the complex function e^z and its associated geometry suddenly mean nothing to you. Suppose that an analytic continuation of e^x to complex values exists, and call it $E(z)$. We can now show that E must be subject to precisely the same law, and without even knowing what E is!

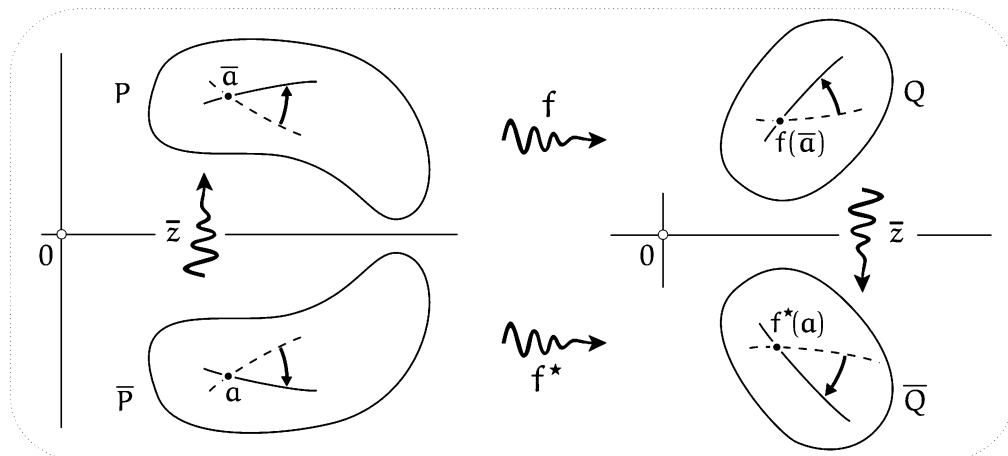
Let $F_\zeta(z) \equiv E(\zeta) \cdot E(z)$, and let $G_\zeta(z) \equiv E(\zeta+z)$. First note that for fixed ζ both $F_\zeta(z)$ and $G_\zeta(z)$ are analytic functions of z . Now suppose that ζ is real, so that $E(\zeta) = e^\zeta$. If z now moves on a segment of the real axis then it follows from the real identity that $F_\zeta(z) = G_\zeta(z)$; but from our recent results we know this implies that they are equal everywhere. If we hold z fixed instead, then analogous reasoning yields $F_\zeta = G_\zeta$, and we conclude that

$$E(\zeta) \cdot E(z) = E(\zeta + z)$$

for complex values of both ζ and z . It should be clear that this reasoning extends to any identity, even one involving more than two variables.

5.11.5 Analytic Continuation via Reflections

Quite distinct from questions of existence and uniqueness is the problem of actually *finding* an analytic continuation. The above ideas and results are mute on this



[5.28] Extending an analytic mapping via reflection. If the analytic function f maps P to Q , then we can use it to construct an analytic mapping from $\bar{P}$ to $\bar{Q}$, given by $f^*(z) \equiv \overline{f(\bar{z})}$. As we see, the two reflections reverse angles twice, resulting in a conformal mapping.

issue, although it could reasonably be claimed that the persistence of identities is a practical help. We next explain a *Symmetry Principle* (due to Schwarz) which enables one to find a continuation easily and explicitly, albeit under rather special circumstances.

We first describe how it is possible to use two reflections to construct a new analytic function from an old one. Suppose an analytic function f is defined on a region P , the image of which is Q (see [5.28]). Let $\bar{P}$ and $\bar{Q}$ be the reflections of these regions across the real axis. We can now use f to construct an analytic mapping from $\bar{P}$ to $\bar{Q}$, namely

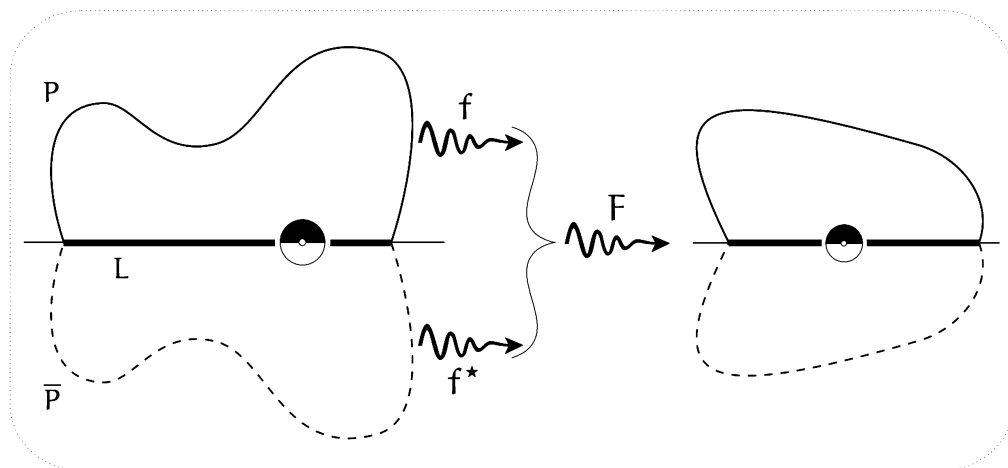
$$f^*(z) \equiv \overline{f(\bar{z})}.$$

The figure explains why f^* is conformal, and hence analytic. All three stages, $a \mapsto \bar{a} \mapsto f(\bar{a}) \mapsto \overline{f(\bar{a})}$, preserve the magnitude of an angle at a ; the first reflection reverses the sense, then f preserves the reversed sense, and finally the second reflection undoes the damage, restoring the angle to pristine condition at $f^*(a)$.

In general this mapping f^* will not be a continuation of f in any sense, rather it is an entirely new mapping. This should become clear if you imagine moving P downwards until some of it crosses the real axis. P and $\bar{P}$ now overlap so that $P \cap \bar{P}$ constitutes a common domain for f and f^* , but we hope you can see that there is no reason for them to agree with each other. This is clearer still if we take an example:

$$f = (\text{rotation of } \phi) \xrightarrow{\text{exercise}} f^* = (\text{rotation of } -\phi).$$

Although it is generally not a continuation of f , this new mapping f^* (together with its soon to be introduced generalization to circles) is very useful in its own right. In



[5.29] Analytic continuation via reflection. Suppose that the analytic function f defined in P is the complex extension of a real function defined on L . Then the function $f^*(z) \equiv \overline{f(\bar{z})}$ introduced in the previous figure is the analytic continuation of f into $\bar{P}$. Generalizing, if f maps a line-segment L (not necessarily real) to another line-segment $\hat{L}$, then we can analytically continue from one side of L to the other by using the fact that points symmetric in L map to points symmetric in $\hat{L}$.

Chapter 12 we will show that it is intimately connected with the so-called *method of images* of electrostatics and fluid dynamics.

We now turn to the special circumstance under which f^* is the analytic continuation of f . Suppose that f is itself the complex generalization of a real function, and let P have a part L of its boundary along the real axis, as in [5.29]. Since f is real on L , the image set Q will also border on the real axis. Unlike the general situation previously considered, f and f^* will now automatically agree on their common domain $P \cap \bar{P} = L$, for if z is real then

$$f^*(z) = \overline{f(\bar{z})} = \overline{f(z)} = f(z).$$

We can now think of f and f^* as being two parts of a single analytic mapping F on $P \cup \bar{P}$. Indeed, by considering what happens to the two halves of an infinitesimal disc centred on L , it's clear that F is analytic there, for the image is another infinitesimal disc. [What happens if we are at a critical point?] Once again, notice how different this is from the case of real functions, for we could easily join two pieces of graph together with a kink at the join; their values would then agree, while their derivatives would not.

Of course if f is already defined in $\bar{P}$ (as well as P) then f^* must simply reproduce the mapping that's already there. For example, the formula for the complex generalization $\sin z$ is valid everywhere, so it should be subject to the symmetry $f^*(z) = f(z)$. Indeed if we follow the three steps of $a \mapsto f^*(a)$ then we do find that

$$a \mapsto \bar{a} \mapsto \frac{e^{i\bar{a}} - e^{-i\bar{a}}}{2i} \mapsto \frac{e^{-ia} - e^{ia}}{-2i} = \sin a.$$

We can rephrase our result in a more symmetric and slightly generalized [exercise] form. If f maps a line-segment L (not necessarily real) to another line-segment $\widehat{L}$, then we can analytically continue from one side of L to the other by using the fact that *points symmetric in L map to points symmetric in $\widehat{L}$* .

This sounds very reminiscent of the conservation of symmetry by Möbius transformations that we discovered in Chapter 3, and indeed by fusing these two symmetry principles we can obtain a significant generalization of our result. Suppose that instead of mapping a particular⁹ line to a line, f sends a part C of a circle to a part $\widehat{C}$ of another circle. We can reduce this to the previous case by using two Möbius transformations to send $C \mapsto L$, and $\widehat{C} \mapsto \widehat{L}$. We deduce that *points symmetric in C map to points symmetric in $\widehat{C}$* .

As a mixed example, imagine that f maps part of the unit circle to part of the real axis. If f is only known inside the circle then the above result tells us [exercise] that there is an analytic continuation to the exterior given by

$$f^\dagger(z) \equiv \overline{f\left(\frac{1}{\bar{z}}\right)}.$$

The complete analytic function F is then defined to be f inside the circle, and $f^\dagger$ outside the circle. By construction, this function sends symmetric pairs of points to conjugate images: $F^\dagger(z) = \overline{F(z)}$.

Using what is now known as *Schwarzian reflection*¹⁰, in 1870 H. A. Schwarz was able to generalize his Reflection Principle beyond lines and circles to more general curves. We end this chapter with a description of Schwarz's simple, yet fascinating, idea. The key is to use an analytic function to fake conjugation.

We know that reflecting every point across the real axis ($z \mapsto \bar{z}$) is not an analytic function. However, given a sufficiently smooth¹¹ curve K , it is possible to find an analytic function $\mathcal{S}_K(z)$ that *selectively* sends just the points of K to their conjugates:

$$z \in K \implies \mathcal{S}_K(z) = \bar{z}.$$

Davis and Pollak (1958) christened $\mathcal{S}_K$ the *Schwarz function* of K . We can now define the Schwarzian reflection of z across K to be $\tilde{z} = \mathcal{R}_K(z)$, where

$$\mathcal{R}_K(z) \equiv \overline{\mathcal{S}_K(z)}.$$

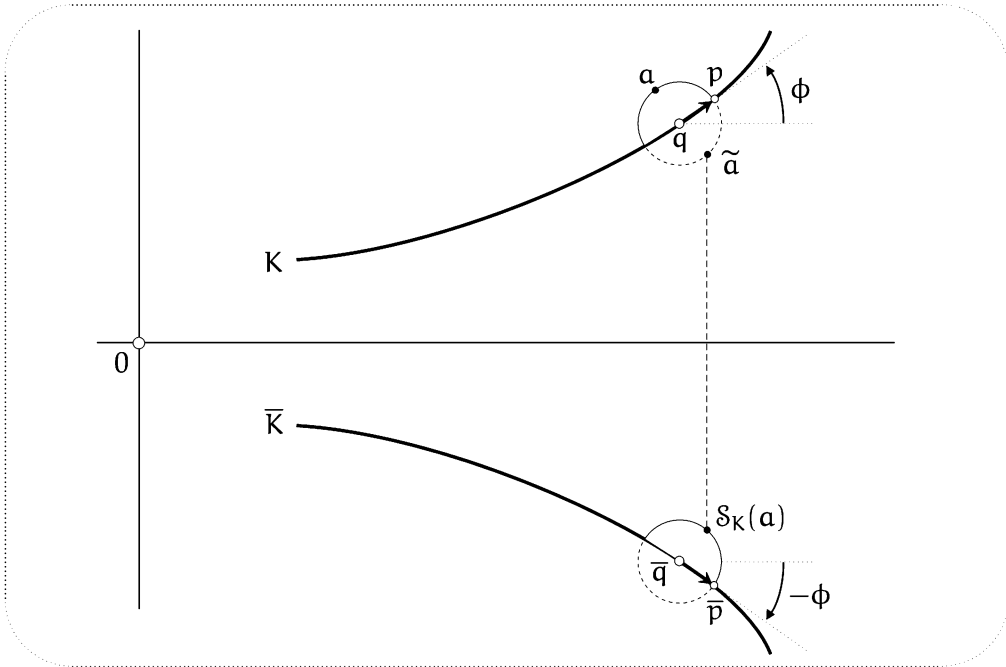
To see why this is a good idea, consider [5.30]. First note that points on K are unaffected, in accord with the ordinary notion of reflection, e.g.,

$$\tilde{q} = \overline{\mathcal{S}_K(q)} = \overline{(\bar{q})} = q.$$

⁹ We stress "particular", because if a general line were sent to a line then the mapping could only be linear. Similarly, in the new case, if a general circle were sent to a circle then the mapping would have to be a Möbius transformation.

¹⁰ Schwarz (1972a).

¹¹ The curve must in fact be "analytic". On this point, see Davis (1974), which also contains many interesting applications of the Schwarz function. A more advanced work is Shapiro (1992).



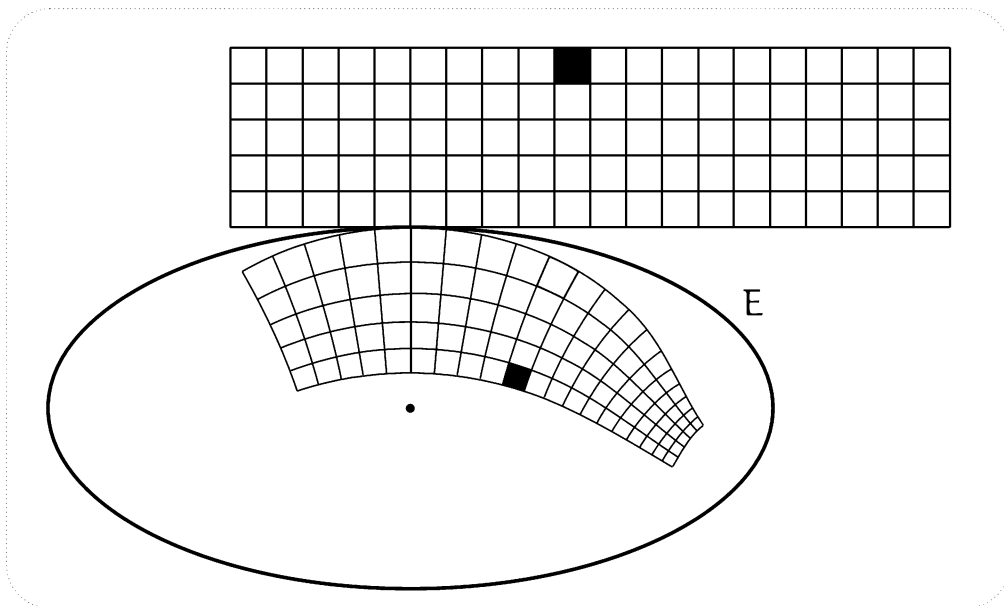
[5.30] Schwarzian Reflection. The Schwarz function S_K of a curve K is defined to be the analytic mapping that mimics conjugation on K : if z lies on K , then $S_K(z) = \bar{z}$. We can then define Schwarzian reflection across K to be $z \mapsto \tilde{z} = \Re_K(z) \equiv \overline{S_K(z)}$. As we see, very close to K , Schwarzian reflection becomes ordinary reflection. If K is a line or circle, then Schwarzian reflection is precisely reflection/inversion in K .

Next, observe that since S_K is analytic, an infinitesimal disc centred at q is amplitwisted (not reflected) to a disc centred at $\bar{q}$. Furthermore, by noting how $\overrightarrow{qp}$ is mapped to $\overrightarrow{\bar{q}\bar{p}}$, it follows that on K

$$\text{amplification} = 1 \quad \text{and} \quad \text{twist} = -2\phi \quad \implies \quad S'_K = e^{-i2\phi},$$

where ϕ is the angle that the tangent to K makes with the horizontal. It is now clear from the symmetry of the figure that if the point a is on the infinitesimal circle, then $\tilde{a}$ is indeed its reflection across the tangent of K . Thus, at least very close to K , $z \mapsto \tilde{z}$ is a reasonable generalization of the reflection concept. Furthermore, reflecting in K twice yields the identity mapping, as it should. For since $\Re_K$ is anticonformal, $\Re_K \circ \Re_K$ is conformal, i.e., analytic. But since this function maps each point of K to itself, and since an analytic function is determined by its values on a curve, $\Re_K \circ \Re_K$ must be the identity mapping.

We leave it to the exercises for you to show that if K is a line or a circle then $\tilde{z}$ is just the ordinary reflection, even if z is far from K . For example, the unit circle C may be written as $\bar{z}z = |z|^2 = 1$, so that on C we have $\bar{z} = (1/z)$. Thus its Schwarz function is $S_C(z) = (1/z)$, and so $\Re_C(z) = (1/\bar{z})$, which is just inversion in C .



[5.31] Schwarzian reflection in an ellipse.

Let us give a less trivial example, namely, reflection in the ellipse E with equation $(x/a)^2 + (y/b)^2 = 1$. Writing $x = \frac{1}{2}(z + \bar{z})$ and $y = \frac{1}{2i}(z - \bar{z})$, then solving for $\bar{z}$ in terms of z , we find [exercise] that

$$s_E(z) = \frac{1}{a^2 - b^2} \left[(a^2 + b^2)z - 2ab \sqrt{z^2 + b^2 - a^2} \right].$$

With $a = 2$ and $b = 1$, for example, Schwarzian reflection is given by

$$\mathfrak{R}_E(z) = \frac{1}{3} \left[5\bar{z} - 4\sqrt{\bar{z}^2 - 3} \right],$$

which is illustrated in [5.31]. We encourage you to verify this figure with your computer, as well as to examine the effect of $\mathfrak{R}_E$ on other shapes.

With the proper concept of reflection in hand, we may now generalize the above method of analytic continuation across lines and circles to more general curves. Let f be an analytic mapping defined in a region P bordering on a curve L that is smooth enough to possess a Schwarz function, and let $\hat{L} = f(L)$ be the image of L under f . Much as in figures [5.28] and [5.29], we may now analytically continue f across L by demanding that points that are symmetric in L map to points that are symmetric in $\hat{L}$. Thus [draw a picture!] the continuation $f^\dagger$ of f to $\tilde{P} = \mathfrak{R}_L(P)$ is

$$f^\dagger = \mathfrak{R}_{\hat{L}} \circ f \circ \mathfrak{R}_L.$$

By the same argument as in [5.28], this is indeed analytic in $\tilde{P}$, for it is the composition of one conformal mapping with two anticonformal mappings. Also, $f^\dagger = f$ on L . The complete analytic function F given by f in P and $f^\dagger$ in $\tilde{P}$ is then subject to the symmetry $F^\dagger = F$. This is *Schwarz's Symmetry Principle*.

Our previous results are just special cases of this construction. For example, if L and $\widehat{L}$ are segments of the real line then $\Re_L(z) = \Re_{\widehat{L}}(z) = \bar{z}$, so $f^\ddagger = f^*$, as before. Similarly, if L is an arc of the unit circle (so that $\Re_L(z) = 1/\bar{z}$) and $\widehat{L}$ is a segment of the real line (so that $\Re_{\widehat{L}} = \bar{z}$), then $f^\ddagger = f^\dagger$, as before.

5.12 Exercises

- 1 Show that if $f = u + iv$ is analytic then $(\nabla u) \cdot (\nabla v) = 0$, where ∇ is the “gradient operator” of vector calculus. Explain this geometrically.
- 2 Show that the real and imaginary parts of an analytic function are *harmonic*, i.e., they both automatically satisfy *Laplace’s equation*:

$$\Delta \Phi = 0,$$

where Δ (which is often instead written ∇^2) is defined by $\Delta \equiv \partial_x^2 + \partial_y^2$ and is called the *Laplacian*. [In Chapter 12 we will see that this equation represents a crucial link between analytic functions and physics.]

- 3 Use the previous exercise (*not* calculation) to show that each of the following is “harmonic”.
 - (i) $e^x \cos y$.
 - (ii) $e^{(x^2-y^2)} \cos 2xy$.
 - (iii) $\ln |f(z)|$, where $f(z)$ is analytic.
- 4 What is the most general function $u = ax^2 + bxy + cy^2$ that is the real part of an analytic function? Construct this analytic function, and express it in terms of z .
- 5 Which of the following are analytic?
 - (i) $e^{-y} (\cos x + i \sin x)$.
 - (ii) $\cos x - i \sin y$.
 - (iii) $r^3 + i3\theta$.
 - (iv) $[r e^{r \cos \theta}] e^{i(\theta + r \cos \theta)}$.
- 6 Solve the Polar CR equations given that $\partial_\theta v \equiv 0$. Express your answer in terms of a familiar function, and interpret everything you have done geometrically.
- 7 Use the Cartesian CR equations to show that the *only* analytic mapping that sends parallel lines to parallel lines is the linear mapping. [Hint: Begin with the case of horizontal lines being mapped to horizontal lines. How does this translate into ‘Equationspeak’? Now solve CR.]
- 8 Calculate, then draw on a picture, a possible location for $\log(1 + i)$. Draw a small shape at $1 + i$. Use the amplitwist of $\log(z)$ to draw its image. Verify this using your computer.
- 9 Derive the quotient rule in an analogous way to the product rule (see page 256). [Hint: Multiply top and bottom of (A/B) by $(b - \xi g')$.]
- 10 Consider the polynomial $P(z) = (z - a_1)(z - a_2) \dots (z - a_n)$.

- (i) Show that the critical points of $P(z)$ are the solutions of

$$\frac{1}{z - a_1} + \frac{1}{z - a_2} + \cdots + \frac{1}{z - a_n} = 0.$$

- (ii) Let K be a circle with centre p . By considering the conjugate of the equation in (i), deduce that p is a critical point if and only if it is the centre of mass of the inverted points $\mathcal{I}_K(a_j)$.

- (iii) Show that the equation in (i) is equivalent to

$$\frac{z - a_1}{|z - a_1|^2} + \frac{z - a_2}{|z - a_2|^2} + \cdots + \frac{z - a_n}{|z - a_n|^2} = 0,$$

and by interpreting the LHS as a (positively) weighted sum of the vectors from z to the roots of $P(z)$, deduce *Lucas' Theorem: The critical points of a polynomial in $\mathbb{C}$ must all lie within the convex hull of its zeros.* This is a complex generalization of Rolle's Theorem in ordinary calculus. [Hint: Use the fact that (2.32) on page 116 is still valid even if the masses are not equal.]

- 11 Use $(e^z)' = e^z$ to show that the derivatives of all the trig functions are given by the familiar rules of real analysis.

- 12 Provided it is properly interpreted, show that $(z^\mu)' = \mu z^{\mu-1}$ is still true even if μ is complex.

- 13 (i) If a is an arbitrary constant, show that the series

$$f(z) = 1 + az + \frac{a(a-1)}{2!}z^2 + \frac{a(a-1)(a-2)}{3!}z^3 + \cdots$$

converges inside the unit circle.

- (ii) Show that $(1+z)f' = af$.

- (iii) Deduce that $[(1+z)^{-a}f]' = 0$.

- (iv) Conclude that $f(z) = (1+z)^a$.

- 14 As we pointed out in Chapter 3, stereographic projection has a very practical use in drawing a conformal map of the world. Once we have this map we can go on to generate further conformal maps, simply by applying different analytic functions to it. One particularly useful one was discovered (using other means) by Gerhard Mercator in 1569. We can describe it (though he could not have) as the result of applying $\log(z)$ to the stereographic map.

- (i) Look up both a stereographic map and a Mercator map in an atlas, and make sure you can relate the changes in shape you see to your understanding of the complex logarithm.
- (ii) Imagine plotting a straight-line course on a Mercator map and then actually travelling it on the high seas. Show that as you sail, the reading of your compass never changes.

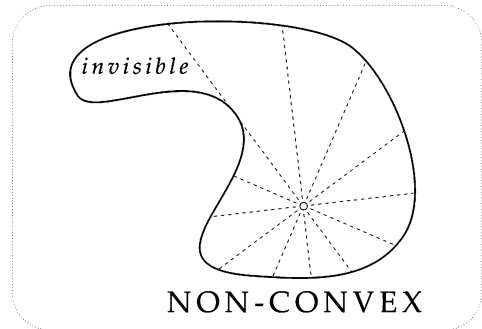
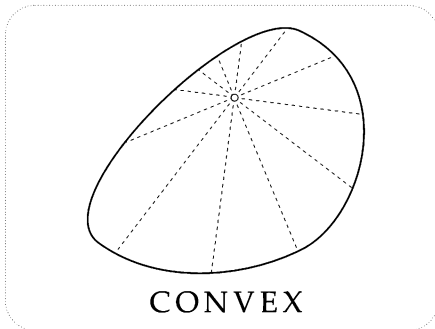
- 15 (i) By noting that the unit tangent (in the counterclockwise direction) to an origin-centred circle can be written as $\hat{\xi} = i(z/|z|)$, show that formula (5.24) for the curvature of the image of such a circle can be written as

$$\tilde{\kappa} = \frac{1 + \operatorname{Re} \left[\frac{z f''}{f'} \right]}{|z f'|}.$$

- (ii) What should this formula yield if $f(z) = \log z$? Check that it does.
- (iii) What should this formula yield if $f(z) = z^m$? Check that it does. What is the significance of the negative value of $\tilde{\kappa}$ when m is negative? [Hint: Which way does the velocity complex number of the image rotate as z travels counterclockwise round the original circle?]
- 16 As illustrated below, a region is called *convex* if all of it is visible from an arbitrary vantage point inside. Let an analytic mapping f act on an origin-centred circle C to produce a simple image curve $f(C)$, the interior of which is convex.
- (i) From the formula of Ex. 15, deduce that if f maps the interior of C to the interior of $f(C)$, then the following inequality holds at all points z of C :

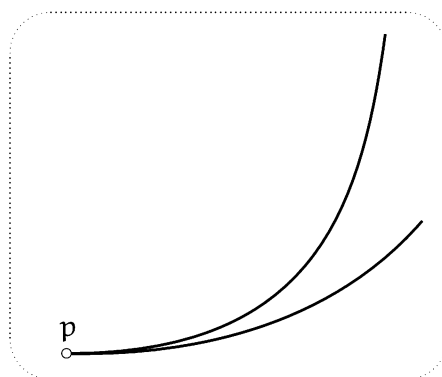
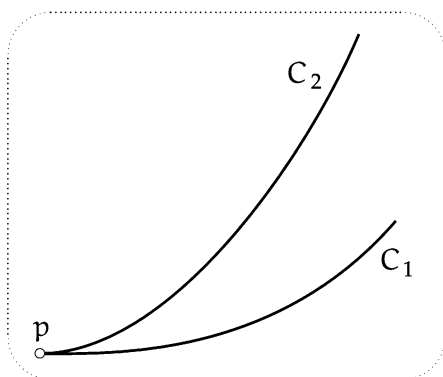
$$\operatorname{Re} \left[\frac{z f''}{f'} \right] \geq -1.$$

- (ii) What is the analogous inequality when f maps the interior of C to the exterior of $f(C)$. [Hint: Ex. 4, p. 239.]



- 17 Let S be a directed line-segment through a point $p = x + iy$.
- (i) Let $f(z) = e^z$. Without calculation, decide which direction of S yields an image $f(S)$ having vanishing curvature at $f(p)$.
- (ii) The complex curvature $\mathcal{K}$ must therefore point in one of the two orthogonal directions. Which? By considering the image of S when it points in this direction, deduce the value of $|\mathcal{K}|$, and thereby conclude that $\mathcal{K}(p) = ie^{-x}$.
- (iii) Use (5.29) to verify this formula.

- (iv) Repeat as much as possible of the above analysis in the cases $f(z) = \log(z)$ and $f(z) = z^m$, where m is a positive integer. [In neither of these cases will you be able to see the exact value of $|\mathcal{K}(p)|$.]
- (v) According to the geometric reasoning in Ex. 18, p. 241, the amplification of a Möbius transformation $M(z) = \frac{az+b}{cz+d}$ is constant on each circle centred at $-(d/c)$. Thus the complex curvature of M should be tangent to these concentric circles. Verify this by calculating $\mathcal{K}$.
- (vi) Use a computer to verify figure [5.21] for all four mappings above.
- 18 Let two curves C_1 and C_2 emerge from a point p in the same direction. Two examples are illustrated below.



Although in both cases the angle at p is zero, there is a great temptation to say that the curves on the right meet at a smaller “angle” than those on the left. Any putative definition of such an “angle” Θ should (presumably) be *conformally invariant*: if the curves are mapped to $\tilde{C}_1$ and $\tilde{C}_2$ by a mapping f that preserves ordinary angles (i.e., an analytic mapping), then the new “angle” $\tilde{\Theta}$ should equal the old “angle” Θ .

- (i) [Newton \(1670\)](#) attempted to define such a Θ as the difference of the curvatures of C_1 and C_2 at p : $\Theta \equiv \kappa_1 - \kappa_2$. Use (5.31) to show that this definition is not quite conformally invariant: $\tilde{\Theta} = \Theta/|f'(p)|$.
- (ii) Consider an infinitesimal disc D (radius ϵ) centred at p . Let c_1 and c_2 be the centres of curvature of C_1 and C_2 , and let $\mathcal{D}$ be the difference between the angular sizes of D as seen from c_1 and c_2 . Show that $\mathcal{D} = \epsilon\Theta$. If a conformal mapping f is applied to C_1 , C_2 , and D , deduce that $\tilde{\mathcal{D}} = \mathcal{D}$. [Of course this is not what we were after: $\mathcal{D}$ is (a) infinitesimal, and (b) not defined by the curves alone. The discovery of a true conformal invariant had to await [Kasner \(1912\)](#). See Ex. 10, p. 649.]

- 19 In more advanced work on Möbius transformations (e.g., [Nehari \(1952\)](#) and [Beardon \(1984\)](#)), an important role is played by the so-called *Schwarzian derivative* $\{f(z), z\}$ of an analytic function $f(z)$ with respect to z :

$$\{f(z), z\} \equiv \left(\frac{f''}{f'} \right)' - \frac{1}{2} \left(\frac{f''}{f'} \right)^2.$$

- (i) Show that the Schwarzian derivative may also be written as

$$\{f(z), z\} = \frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'} \right)^2.$$

- (ii) Show that $\{az + b, z\} = 0 = \{(1/z), z\}$.
 (iii) Let f and g be analytic functions, and write $w = f(z)$. Show that the Schwarzian derivative of the composite function $g[f(z)] = g[w]$ is given by the following “chain rule”:

$$\{g(w), z\} = [f'(z)]^2 \{g(w), w\} + \{f(z), z\}.$$

- (iv) Use the previous two parts to show that all Möbius transformations have vanishing Schwarzian derivative. [*Hint*: Recall that the mappings in part (ii) generate (via composition) the set of all Möbius transformations.] *Remark*: Ex. 19, p. 481 shows that the converse is also true: If $\{f(z), z\} = 0$ then $f = \text{Möbius}$. Thus Möbius transformations are completely characterized by their vanishing Schwarzian derivative.
 (v) Use the previous two parts to show that the Schwarzian derivative is “invariant under Möbius transformations”, in the following sense: if M is a Möbius transformation, and f is analytic, then

$$\{M[f(z)], z\} = \{f(z), z\}.$$

- 20 Think of the real axis as representing time t , and let a moving particle $w = f(t)$ (where $f(z)$ is analytic) trace an orbit curve C . The velocity is then $v = \dot{w} = f'(t)$.

- (i) Use (5.24) to show that the curvature of C is

$$\tilde{\kappa} = \frac{\text{Im}(\dot{v}/v)}{|v|}.$$

- (ii) Argue that this result does not in fact depend on C being produced by an analytic mapping, but is instead true of any motion for which the velocity v and acceleration $\dot{v}$ are well-defined.
 (iii) Show that the formula may be rewritten as

$$\tilde{\kappa} = \frac{\text{Im}(\bar{v}\dot{v})}{|v|^3}.$$

(iv) Deduce that it may also be written vectorially as

$$\tilde{\kappa} = \frac{|\mathbf{v} \times \dot{\mathbf{v}}|}{|\mathbf{v}|^3}.$$

By considering C to be the “osculating plane”—see [Needham \(2021\)](#)—of a curve in 3-dimensional space, we see that this formula holds in that case also.

21 In 3-dimensional space, let (X, Y, Z) be the coordinates of a moving particle. If $X = a \cos \omega t$, $Y = a \sin \omega t$, $Z = bt$, then the path traced by the particle is a helix.

- (i) Give interpretations for the numbers a , ω , and b .
- (ii) If a and ω remain fixed, what does the helix look like in the two limiting cases of b becoming very small or very large? What if a and b remain fixed while ω becomes very small or very large?
- (iii) What limiting values would you anticipate for the curvature of the helix for each of the limiting cases considered in (ii)?
- (iv) Use Ex. 20(iv) to show that the curvature of the helix is

$$\tilde{\kappa} = \frac{a \omega^2}{b^2 + a^2 \omega^2},$$

and use this to confirm your hunches in (iii).

22 Continuing from Ex. 20, take $f(z)$ to be a general Möbius transformation:

$$w = f(t) = \frac{at + b}{ct + d},$$

where $\Delta \equiv (ad - bc) \neq 0$. Show that the curvature of this path is

$$\tilde{\kappa} = \left| \left(\frac{2c^2}{\Delta} \right) \operatorname{Im} \left(\frac{d}{c} \right) \right|,$$

in agreement with Ex. 18, p. 210. The fact that this is constant provides a new proof that the image is a circle, for only circles have constant curvature.

23 As another continuation of Ex. 20, let us see how the Schwarzian derivative $\{f(z), z\}$ of Ex. 19 arises rather naturally in the context of curvature.

(i) Show that

$$\frac{d\tilde{\kappa}}{dt} = \frac{1}{|f'|} \operatorname{Im}\{f(z), z\}.$$

[This formula was discovered by G. Pick. For elegant applications, see [Beardon \(1987\)](#). For another connection between curvature and Schwarzian derivatives, see Ex. 28(iii).]

(ii) Use Ex. 19, part (iv) to deduce that if $f(z)$ is a Möbius transformation then $\frac{d}{dt} \tilde{\kappa} = 0$. Why is this result geometrically obvious?

24 Let the position at time t of a moving particle in $\mathbb{C}$ be $z(t) = r(t) e^{i\theta(t)}$.

(i) Show that the acceleration of the particle is

$$\ddot{z} = [\ddot{r} - r\dot{\theta}^2] e^{i\theta} + [2\dot{r}\dot{\theta} + r\ddot{\theta}] ie^{i\theta}.$$

(ii) What are the radial and transverse components of the acceleration?

(iii) If the particle is moving in a central force field, with the centre of force at the origin, deduce that the areal speed $\mathcal{A} = (r^2\dot{\theta}/2)$ is constant. For a beautiful geometric proof of this fact, see [Newton \(1687, Prop. 1\)](#), a comic-strip rendition of which may be found in [Needham \(2021, §11.7.3\)](#).

25 Sometimes the circle of convergence of a power series is so densely packed with singularities that it becomes a genuine barrier for the geometric mapping, beyond which it cannot be continued. This is called a *natural boundary*. An example of this is furnished by

$$f(z) = z + z^2 + z^4 + z^8 + z^{16} + \dots,$$

which converges inside the unit circle. Show that every point of $|z| = 1$ is either a singularity itself, or else has singularities arbitrarily near to it. [*Hint*: What is $f(1)$? Now note that $f(z) = z + f(z^2)$, and deduce that f is singular when $z^2 = 1$. Continuing in this manner, show that the 2^n -th roots of unity are all singular.]

26 Unlike inversion in a circle, show that Schwarzian reflection in an ellipse E (see figure [5.31]) does not interchange the interior and the exterior. Indeed, how does $\Re_E(z)$ behave for large values of $|z|$?

27 (i) If L is a line passing through the real point X , and making an angle σ with the horizontal, then show that its Schwarz function is

$$\mathcal{S}_L(z) = z e^{-i2\sigma} + X(1 - e^{-i2\sigma}).$$

(ii) If C is a circle with centre p and radius r , show that its Schwarz function is

$$\mathcal{S}_C(z) = \bar{p} + \frac{r^2}{z - p}.$$

(iii) Verify the claim that in both these cases $z \mapsto \Re(z)$ is the ordinary reflection, even if z is far from the curve.

28 Let a be a point on a (directed) curve K having Schwarz function $\mathcal{S}(z)$.

(i) Show that the curvature of K at a is

$$\kappa \equiv \dot{\phi} = \frac{i}{2} \cdot \frac{\mathcal{S}''(a)}{[\mathcal{S}'(a)]^{3/2}},$$

where ϕ is the angle in [5.30], and the dot denotes differentiation with respect to distance l along K (in the given sense). Deduce that

$$|\kappa| = |\mathcal{S}''/2|.$$

[Hints: Since $\mathcal{S}$ is analytic, so is $\mathcal{S}'$. Thus to calculate $\mathcal{S}'' = d\mathcal{S}'/dz$ we need only find the change $d\mathcal{S}'$ in $\mathcal{S}'$ produced by an infinitesimal movement dz of z , taken in any one direction of our choosing. At a let us choose dz along K , so that $dz = e^{i\phi} dl$. The corresponding change in $\mathcal{S}'$ is then determined solely by the shape of K , for the values of $\mathcal{S}'$ on K are given by $\mathcal{S}' = e^{-2i\phi}$.]

- (ii) Deduce that the centre of curvature of K at a is $\{a + 2[\mathcal{S}'(a)/\mathcal{S}''(a)]\}$.
- (iii) Show that the rate of change of the curvature of K is given by the “Schwarzian derivative” [Ex. 19] of the Schwarz function:

$$\dot{\kappa} = \frac{i}{2\mathcal{S}'} \{\mathcal{S}(z), z\}.$$

- 29 Check the result of Ex. 28 (i) by applying it to the results of Ex. 27.
- 30 Let a be a point on a curve K having Schwarz function $\mathcal{S}(z)$. By the still unproven result on the infinite differentiability of analytic functions, $\mathcal{S}(z)$ may be expanded into a Taylor series in the vicinity of a :

$$\mathcal{S}(z) = \mathcal{S}(a) + \mathcal{S}'(a)(z-a) + \frac{1}{2!}\mathcal{S}''(a)(z-a)^2 + \frac{1}{3!}\mathcal{S}'''(a)(z-a)^3 + \dots$$

- (i) Show that the Schwarz function of the tangent line to K at a is given by the first two terms of the series above. This reconfirms something we saw in [5.30]: very close to a , reflection in the tangent is a good approximation to Schwarzian reflection.
- (ii) It is natural to suspect that a better approximation to $\mathfrak{R}_K(z)$ would be inversion in the circle of curvature (call it C) of K at a . Let’s verify this. Use Ex. 28(ii) and Ex. 27(ii) to find $\mathcal{S}_C$, and show that it may be written as

$$\mathcal{S}_C(z) = \bar{a} + \frac{2\overline{\mathcal{S}'}}{\overline{\mathcal{S}''}} + \frac{2(\mathcal{S}')^2}{\mathcal{S}''} \left[1 - \frac{\mathcal{S}''}{2\mathcal{S}'}(z-a) \right]^{-1},$$

where it is understood that the derivatives are all evaluated at a . Show that the first three terms in the binomial expansion of $\mathcal{S}_C$ agree with those of $\mathcal{S}$, but that they generally differ thereafter. [Hint: You will need the fact that $(\overline{\mathcal{S}'}/\overline{\mathcal{S}''}) = -(\mathcal{S}')^2/\mathcal{S}''$ on K . Prove this.]

- (iii) If the curvature κ of K were constant then K would be identical to its circle of curvature. The fact that $\mathcal{S}$ and $\mathcal{S}_C$ disagree beyond the third term thus reflects the fact that κ does change. One is thus led to guess that the faster κ changes, the greater the discrepancy between $\mathfrak{R}_K$ and inversion in C . Continuing from the last part, use Ex. 28(iii) to verify this hunch in the following precise form:

$$\mathcal{S}_C(z) - \mathcal{S}(z) \approx (i/3)[\mathcal{S}']^2 \dot{\kappa} (z-a)^3.$$

- 31 Let C and D be intersecting circles. Let us say that “ D is symmetric in C ” if reflection (inversion) in C maps D into itself. We know this occurs if and only if D is orthogonal to C , so

$$D \text{ is symmetric in } C \iff C \text{ is symmetric in } D.$$

Briefly, we may simply say that “ C and D are symmetric”. Let’s see what happens if we generalize C and D to intersecting arcs possessing Schwarz functions, and generalize inversion to Schwarzian reflection.

- (i) Explain why the statement “ D is symmetric in C ” is the same as “if the point d lies on D then $\Re_D [\Re_C(d)] = \Re_C(d)$ ”. Must the arcs be orthogonal?
- (ii) If D is symmetric in C , deduce that the mappings $(\Re_D \circ \Re_C)$ and $(\Re_C \circ \Re_D)$ are equal at points of D .
- (iii) Using the fact that these two mappings are analytic (why?), deduce that C must also be symmetric in D . Thus, as with circles, we may simply say that C and D are symmetric.

